Towards human-centered and efficient video synthesis: a survey of multimodal diffusion models

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Received: 4 September 2025 / Accepted: 3 September 2026
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Abstract
Multimodal video diffusion models have emerged as transformative tools for controlled video synthesis, integrating text, images, audio, and pose sequences to generate semantically meaningful content. Despite significant advances, critical gaps persist in temporal consistency, multimodal alignment, and human-centric motion generation. Existing surveys have not addressed clearly the complex interplay between these components, particularly physiological constraints and identity preservation in human motion synthesis. This survey provides a comprehensive analysis through a unified architectural framework, examining spatial-temporal representations and multimodal conditioning mechanisms. We present the first systematic evaluation of human-centric motion modeling, addressing physiological plausibility and identity consistency challenges. Our analysis reveals fundamental trade-offs between computational efficiency and generation quality, with reported specialized techniques such as temporal block pruning achieving up to 523× computational savings under specific baselines with minimal quality degradation (see Sect. 6.3 for comparability caveats). Key findings indicate that current approaches struggle with seamless multimodal integration, human-centric applications face "uncanny valley" effects when physics constraints are too rigid, and identity preservation conflicts with motion dynamics. We introduce MIME-Vid (Multi-modal Integration with Motion Enhancement for Video Generation) as a *conceptual reference framework* that operationalises the unifying principles identified in our analysis (reference-flexibility, physics-perception asymmetry, and hierarchical disentanglement); empirical validation of MIME-Vid is deferred to follow-up work. Furthermore, we propose novel evaluation paradigms and identify future research directions for advancing multimodal video generation.

Keywords Diffusion models · Multimodal synthesis · Video generation · Human-centric AI · Temporal modeling · Generative models · Controllability · Reinforcement learning

1 Introduction

The rapid expansion of video generation technologies, especially diffusion models, has significantly transformed the landscape of content creation and human-computer interaction. While substantial progress has been made in image generation (Ho et al. 2020; Dhariwal and Nichol 2021), the extension to video data introduces additional

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https://doi.org/10.1007/s10462-026-11699-z

Published online: 13 September 2026

Springer

challenges due to the inherent temporal dynamics and high dimensionality (Ho et al. 2022; Yang et al. 2023). These challenges necessitate the development of novel methodologies that can efficiently capture both spatial and temporal correlations while maintaining semantic consistency across extended sequences.

Multimodal video diffusion models have emerged as a promising direction for tackling this complexity. These models, as illustrated in Fig. 1, integrate various forms of input data such as text, image, audio, and pose information to guide the video synthesis process in a more controlled and meaningful manner (Ruan et al. 2023; Wang et al. 2024a). This multimodal approach enables richer interactions and more precise outputs, paving the way for advanced applications in areas such as animation, virtual reality, and video editing.

Despite recent advancements, the field still faces critical issues, including the efficient representation of temporal information, the alignment between modalities, and the control over style and content generation (Guo et al. 2022; Hong et al. 2022). Addressing these problems requires a unified understanding of the architectural designs, training strategies, and evaluation metrics employed across different studies.

The extension from image to video generation amplifies several fundamental challenges that multimodal approaches are uniquely positioned to address:

Temporal Consistency: Unlike static images, videos require maintaining coherent object identities, motion patterns, and visual characteristics across potentially hundreds of frames. The temporal dimension introduces dependencies that must be modeled effectively to avoid artifacts such as flickering, identity drift, or inconsistent motion trajectories (Ho et al. 2022; Esser et al. 2023).

Computational Complexity: Video generation scales quadratically with sequence length in attention mechanisms, creating prohibitive computational requirements for standard diffusion architectures. The challenge intensifies when incorporating multiple input modalities, each requiring specialized processing and cross-modal alignment mechanisms (Zhang et al. 2025; Wu et al. 2023).

Semantic Alignment: Ensuring that generated videos accurately reflect multimodal inputs while maintaining internal consistency represents a significant challenge. This includes aligning textual descriptions with visual

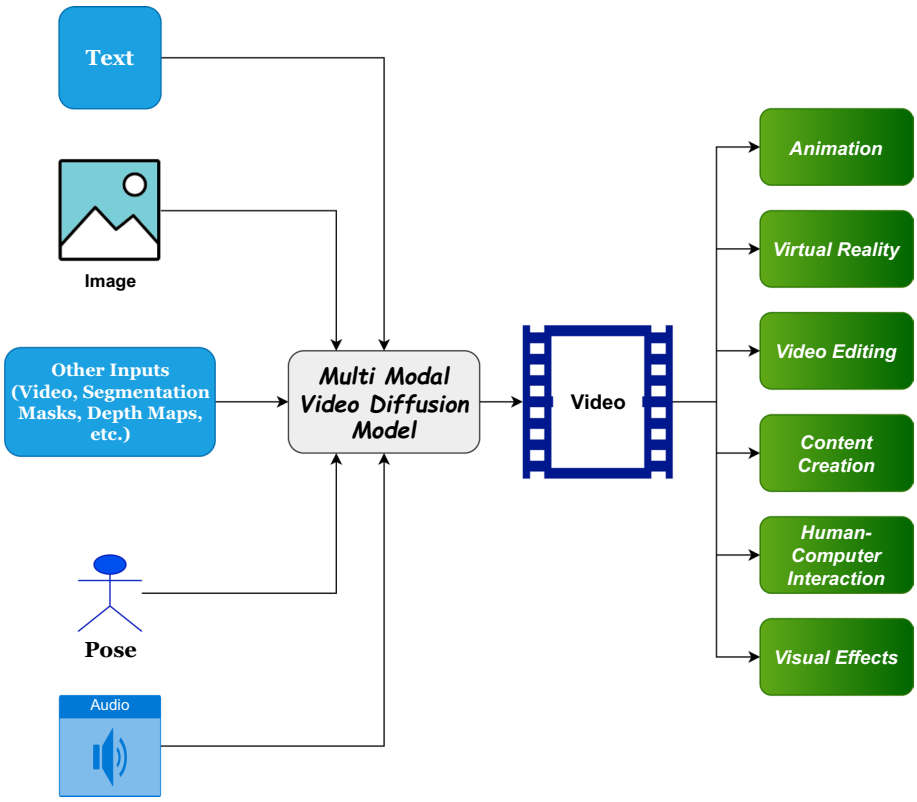


Fig. 1 Conceptual Figure of Multi-Modal Diffusion Models

content, synchronizing audio-visual elements, and preserving identity features from reference images throughout dynamic sequences (Yang et al. 2025; Luo et al. 2023).

Motion Realism: Generating physically plausible and naturally appearing human motion remains particularly challenging, requiring models to understand biomechanical constraints, social interaction patterns, and stylistic variations in movement while maintaining perceptual naturalness (Tevet et al. 2022; Kang et al. 2025; Yuan et al. 2023).

Multimodal video diffusion models offer several key advantages over unimodal approaches. By leveraging complementary information from multiple input sources, these models enable fine-grained control over various aspects of video generation, from high-level semantic content to detailed motion patterns and visual style. Multiple modalities provide redundant and complementary semantic information, leading to more robust and accurate interpretation of user intent and more consistent generation results.

A significant portion of current research focuses on human-centric applications, driven by substantial commercial and creative potential in digital entertainment, virtual reality, education, and social computing. Human-centric video generation presents unique challenges including identity preservation across dynamic sequences, biomechanical plausibility in motion synthesis, accurate representation of social interactions, and nuanced control over movement style and expression. These applications require sophisticated integration of pose sequences, facial landmarks, and appearance features while maintaining temporal consistency and physical realism.

This survey provides a comprehensive examination of multimodal video diffusion models, addressing critical gaps in current literature while highlighting emerging research directions. Beyond summarizing existing literature, this work delivers five contributions that distinguish it from prior surveys: Our key contributions include

1. Unified Theoretical Framework: We present a systematic analysis of foundational principles underlying multimodal video diffusion, examining architectural choices that address temporal modeling and cross-modal alignment challenges. Unlike prior surveys that treat temporal modeling and multi-modal conditioning as separate concerns, we derive a shared noise-prediction formulation (Eq. 9–12) that makes the interplay between the two explicit and reveals where architectural choices trade off.
2. Comprehensive Taxonomy: We develop a structured categorization of existing approaches based on input modalities, architectural designs, and application domains, providing clarity in an increasingly complex research landscape. Our taxonomy uniquely integrates an emerging fifth branch—agent-based and LLM-orchestrated diffusion pipelines (Hong et al. 2024)—that has not been systematically covered by earlier taxonomies.
3. Human-Centric Analysis: We conduct an in-depth examination of human motion generation, identity preservation, and social interaction modeling, identifying key challenges and recent advances in this critical application area. Our analysis yields three unifying principles (reference-flexibility, physics-perception trade-off, and disentanglement granularity; see Sect. 4.2) that explain why current methods fail in systematic rather than ad-hoc ways.
4. Evaluation Framework: We propose a holistic evaluation methodology encompassing quantitative metrics, perceptual assessments, and application-specific benchmarks to guide future research and development. We further contribute a metric-to-application mapping (Table 11) that tells practitioners *which* metrics matter for *which* scenario—a guideline missing from prior evaluation surveys.
5. Research Gap Analysis: We identify critical limitations in current approaches and outline promising directions for advancing the field, including efficiency improvements and emerging applications. We additionally quantify the comparability caveats of reported efficiency gains (Sect. 6.3) to counter overstated claims—an issue we observed across the literature.

The remainder of this survey is organized as follows: Sect. 2 presents the research gap analysis; Sect. 3 establishes the theoretical foundations and examines core architectural components; Sect. 4 focuses on human-centric video generation challenges; Sect. 5 provides a comprehensive literature review categorized by input modalities; Sect. 6 introduces evaluation methodologies with metrics and datasets; Sect. 7 outlines future directions; and

Sect. 8 details the proposed MIME-Vid framework. Through this examination, we aim to provide researchers with both a comprehensive understanding of current capabilities and clear guidance for future innovations in multimodal video diffusion models.

2 Research gap analysis

This section presents a comprehensive analysis of the current landscape in diffusion models for video generation, revealing critical gaps that persist despite the remarkable growth in research output between 2023 and 2026. While numerous surveys have contributed valuable insights to specific aspects of this rapidly evolving field, our systematic examination of existing literature identifies several underexplored intersections that represent significant opportunities for advancement. Through careful analysis of over 100 recent methods and existing survey coverage, we pinpoint three primary areas where current research falls short: the complex interplay between temporal consistency and multi-modal integration, the specialized domain of human-centric motion generation with its unique physiological and aesthetic constraints, and the intricate balance between computational efficiency and generation quality in practical video synthesis applications. These gaps not only highlight the fragmented nature of current research efforts but also underscore the need for a unified framework that addresses these interconnected challenges comprehensively. As shown in Table 1, the reviewed surveys have collectively built a substantial foundation of knowledge, yet several critical gaps remain unaddressed in the current literature. Our comprehnensive analysis reveals that while individual components have received attention, the intersections of key areas remain largely unexplored.

A primary gap emerges at the confluence of temporal consistency and multi-modal integration. While surveys such as (Jiang et al. 2024; Cao et al. 2024), and (Zhang et al. 2024) have thoroughly examined multi-modal aspects, and others like (Xing et al. 2024) and (Melnik et al. 2024) have focused on video generation, the unique challenges arising from their intersection remain inadequately addressed. Recent works such as (Le et al. 2024) have begun to explore this territory, but questions regarding semantic consistency across modalities while maintaining temporal coherence in video generation persist. This gap becomes particularly evident when considering the complex interplay between textual prompts, visual inputs, and temporal dynamics in video synthesis.

The domain of human-centric motion generation represents another significant void in the current literature. While surveys such as (Sun et al. 2024) and (Po et al. 2024) have made valuable contributions to video editing and visual computing respectively, none have specifically tackled the unique challenges inherent in human motion generation. These challenges encompass physiological constraints in motion synthesis, the preservation of natural movement patterns and dynamics, identity consistency across temporal sequences, and the nuanced requirements of motion style transfer and control. The absence of comprehensive coverage in this area is particularly noteworthy given the increasing importance of human-centric applications in video generation.

The efficiency-quality trade-off landscape has received growing attention, as evidenced by recent surveys (Shen et al. 2025; Song et al. 2024), and (Ulhaq and Akhtar 2024). However, these works primarily address general efficiency considerations rather than the specific challenges encountered in video generation. The intricate balance between computational efficiency and temporal consistency, quality considerations in human motion generation, resource optimization for multi-modal integration, and the demands of real-time generation remain underexplored. This gap becomes increasingly significant as the field moves toward practical applications requiring both high-quality output and computational efficiency.

Furthermore, while most existing surveys incorporate evaluation metrics, there is a notable absence of unified frameworks that comprehensively address the multiple dimensions of video generation quality. The field lacks standardized approaches for assessing multi-modal coherence, temporal consistency, human motion naturalness, and perceptual quality in an integrated manner. This gap in evaluation methodology makes it difficult to compare different approaches and assess progress in the field effectively.

https://doi.org/10.1007/s10462-026-11699-z

Table 1 Comprehensive comparison of recent diffusion model surveys (2023-2026)

Survey	Video	Multi-modal	Human Motion	Temp-oral	Effici-ency	Evalu-ation	Vision	Applic-ations	Year
Yang et al. (2023)	✗	✓	✗	✗	✓	✓	✓	General	2023
Cao et al. (2024)	✗	✓	✗	✗	✓	✓	✓	Vision	2024
Croitoru et al. (2023)	✗	✗	✗	✗	✓	✓	✓	Vision	2023
Ulhaq and Akhtar (2024)	✗	✗	✗	✗	✓	✓	✓	Vision	2024
Po et al. (2024)	✓	✓	✗	✓	✗	✓	✓	Visual Computing	2024
Xing et al. (2024)	✓	✗	✗	✓	✗	✓	✓	Video	2024
Zhou et al. (2024)	✓	✓	✗	✓	✗	✓	✓	Video	2024
Melnik et al. (2024)	✓	✗	✗	✓	✗	✓	✓	Video	2024
Zhang et al. (2024)	✗	✓	✗	✗	✗	✓	✓	Content	2024
Sun et al. (2024)	✓	✗	✗	✓	✗	✓	✓	Editing	2024
Chen et al. (2024)	✗	✓	✗	✗	✓	✓	✗	Theory	2024
Mandala (2023)	✗	✓	✗	✗	✓	✓	✓	General	2023
Song et al. (2024)	✗	✗	✗	✗	✓	✓	✓	Efficiency	2024
Jiang et al. (2024)	✗	✓	✗	✗	✗	✓	✓	Control	2024
Le et al. (2024)	✓	✓	✗	✓	✗	✓	✓	Multiple	2024
Shen et al. (2025)	✗	✗	✗	✗	✓	✓	✓	Efficiency	2025
Wu et al. (2025)	✗	✓	✗	✗	✗	✓	✓	Preference	2025
Chen et al. (2025)	✗	✗	✗	✗	✓	✓	✓	Inversion	2025
Ahsan et al. (2024)	✗	✓	✗	✗	✓	✓	✓	Multiple	2024
Chen et al. (2025)	✗	✓	✗	✗	✓	✓	✓	Image	2025
Wang et al. (2026)	✓	✓	✗	✓	✓	✓	✓	Video	2026
Ma et al. (2026)	✓	✓	✗	✓	✗	✓	✓	Control	2026
Wang et al. (2025)	✓	✓	✗	✗	✗	✓	✓	Generative AI	2025
Benjdira et al. (2026)	✓	✓	✗	✗	✓	✓	✓	LLM-Diffusion	2026
Our Survey	✓	✓	✓	✓	✓	✓	✓	General	2026

More recent surveys have further clarified the landscape but continue to leave the human-centric and efficiency dimensions largely unaddressed. Wang et al. (2026) provide a broad, updated review of diffusion-based video generation with dedicated treatment of evaluation metrics, industry solutions, and training engineering, yet without a human-centric focus. Ma et al. (2026) systematically categorize controllable video generation by conditioning signal (single-condition, multi-condition, universal), offering an orthogonal taxonomy to ours that emphasizes the denoising-process control mechanism rather than downstream human-motion fidelity. On the cross-disciplinary front, Wang et al. (2025) survey the unification of multi-modal LLMs and diffusion models, and Benjdira et al. (2026) review progress in fusing the two families. These works underscore the growing importance of LLM-diffusion integration, which our survey situates as one of the enabling substrates for agent-based and instruction-guided video synthesis (see Sect. 2).

This survey aims to address foundational and emerging challenges in multimodal video diffusion models through the following key contributions:

1. Comprehensive Taxonomy and Architectural Analysis: We provide the first systematic categorization of multimodal video diffusion models across input modalities (text, image, pose, audio), temporal modeling approaches (frame-level vs. sequence-level), and architectural innovations (network inflation, grid-based methods, factorized spatio-temporal processing). Our taxonomy encompasses over 100 recent methods, establishing a unified framework for understanding the rapidly evolving landscape of multimodal video generation.
2. Novel MIME-Vid Framework with Motion Enhancement: We introduce MIME-Vid (Multi-modal Integration with Motion Enhancement for Video Generation), a novel framework that integrates advanced Kalman filtering techniques with multi-modal architecture for enhanced temporal consistency and motion realism. This framework addresses critical limitations in existing approaches through physics-informed motion correction, efficient content integration, and post-training quantization for computational optimization.
3. Rigorous Evaluation Methodology and Future Research Roadmap: We establish a comprehensive evaluation framework encompassing video quality assessment (FVD, PSNR, SSIM), motion realism metrics (AED, EPE, TC), and computational efficiency measures (MSR, IT, MUR, QPT). Additionally, we identify critical research gaps in human-centric motion generation, physiological constraints, and multi-person interaction modeling, providing a structured roadmap for advancing the field toward more realistic and controllable video synthesis.

As evidenced by Table 1, our survey uniquely addresses all key aspects while providing comprehensive coverage of both methodological advances and practical applications. This positions our work to fill critical gaps in the current literature and provide valuable insights for future research directions in video generation using diffusion models. By addressing these interconnected challenges comprehensively, we aim to provide researchers and practitioners with a robust foundation for advancing the field of video generation using diffusion models.

The taxonomy of diffusion-based generative models can be structured across several key dimensions, considering computational optimization, modality-specific adaptations, temporal modeling, efficiency-quality trade-offs, and evaluation frameworks. Diffusion models have demonstrated significant advancements in various domains such as image and video generation, speech synthesis, and multi-modal integration (He et al. 2024; Croitoru et al. 2023; Yang et al. 2023). These models balance the efficiency-quality trade-off by leveraging novel architectural improvements, such as structured denoising diffusion models (Austin et al. 2021) and efficient sampling techniques (Ulhaq and Akhtar 2024). Within the realm of video generation, taxonomy categorizes methodologies into frame-level and sequence-level modeling approaches, ensuring temporal consistency and motion synthesis (Xing et al. 2024; Yang et al. 2023). Further refinements, such as human-centric video generation and physiological constraints, enhance applicability in domains requiring high fidelity motion synthesis (Jiang et al. 2024). A crucial dimension in the taxonomy involves multi-modal coherence, integrating text-to-video synthesis and hybrid modal representations (Zhou et al. 2024). Additionally, evaluation frameworks are essential in assessing diffusion models, incorporating unified evaluation metrics that consider perceptual quality, diversity, and computational efficiency (Chen et al. 2024). The taxonomy consolidates the advancements in diffusion models, establishing a foundation for future research in generative AI.

A conceptual diagram of the taxonomy is presented in Fig. 2.

The taxonomy of diffusion-based generative models encompasses four primary branches that collectively address the fundamental challenges in generative modeling. **Computational Optimization** focuses on enhancing model efficiency through structured denoising diffusion processes, efficient sampling techniques, and architectural improvements that reduce computational overhead while maintaining generation quality.

Modality-Specific Adaptations tailors diffusion models to distinct data types, including image generation with frame-level and sequence-level modeling approaches, video generation incorporating temporal consistency and motion synthesis mechanisms, speech synthesis, and multi-modal integration that enables text-to-video synthesis through hybrid modal representations and multi-modal coherence strategies.

Temporal Modeling specifically addresses the unique challenges of sequential data generation, encompassing video generation methodologies that ensure temporal consistency through specialized mechanisms, motion synthesis approaches for realistic movement generation, and human-centric video generation techniques that incorporate physiological constraints.

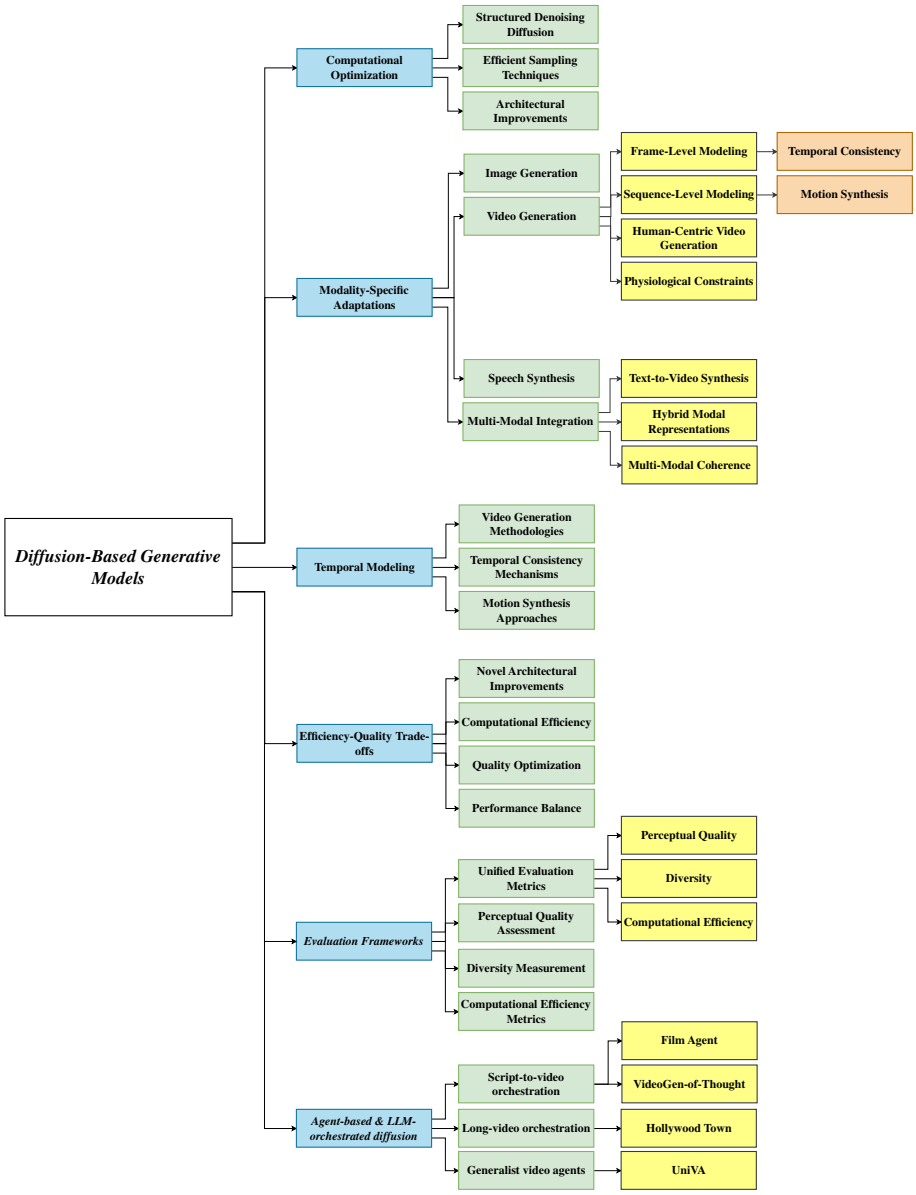


Fig. 2 Taxonomy of research focus and methods in diffusion models

Finally, **Efficiency-Quality Trade-offs** establishes the theoretical and practical frameworks for balancing computational demands with output quality, featuring novel architectural improvements that optimize computational efficiency and quality simultaneously. In addition to the four main branches, there is a comprehensive **Evaluation Frameworks** that assess models through unified evaluation metrics measuring perceptual quality, diversity, and computational efficiency to provide holistic performance assessment across all dimensions of diffusion-based generative modeling.

2.1 Agent-based and LLM-orchestrated diffusion

A fifth, rapidly emerging branch of the taxonomy concerns **agent-based diffusion pipelines**, in which one or more large-language-model (LLM) agents coordinate the planning, decomposition, and iterative refinement of a video generation task rather than treating it as a single end-to-end denoising problem. Unlike the four branches above—which optimize a single model—this branch orchestrates multiple specialized models (and often multiple diffusion backbones) through a collaborative workflow.

The conceptual foundation was established by MetaGPT (Hong et al. 2024), which encodes standardized operating procedures into prompt sequences for multi-agent LLM collaboration and demonstrates that assembly-line decomposition reduces the cascading hallucinations that arise when a single LLM is naively chained. Applied to video synthesis, this paradigm has spawned three distinct sub-families:

- *Script-to-video orchestration*, where one agent writes a screenplay, another plans shots, and diffusion agents render each shot—exemplified by FilmAgent (Xu et al. 2025) and VideoGen-of-Thought (Zheng et al. 2025), which step-by-step generate multi-shot videos with cross-shot consistency.
- *Long-video orchestration*, addressing the fundamental length limitation of single-pass diffusion through cross-modal multi-agent coordination as in Hollywood Town (Wei et al. 2025).
- *Generalist video agents* that unify generation, editing, and understanding under one conversational interface, such as UniVA (Liang et al. 2025), which represents a shift from task-specific diffusion models toward open-ended video tool use.

This branch intersects with the LLM–diffusion fusion trend reviewed by Benjdira et al. (2026) and Wang et al. (2025), and is distinctive in that its contributions are often *system-level* (coordination, memory, planning) rather than *model-level* (architecture, loss, sampling). We note this branch here because its rapid growth suggests that future taxonomies will need to treat orchestration as a first-class design dimension alongside temporal modeling and modality conditioning.

The next section presents the foundation of video diffusion models which will build the theoritical understanding of the various concepts and mathematical techniques starting from basic diffusion models, addressing computing optimization and presenting various modalities used with human focus as well.

3 Foundations of video diffusion models

Recent advancements in generative modeling have witnessed the emergence of diffusion-based approaches as state-of-the-art techniques for image synthesis. These models are now being extended to video generation, presenting both unique challenges and opportunities. This section reviews key literature covering the theoretical foundations, video-specific extensions, and underlying mathematical frameworks of video diffusion models.

3.1 Core theoretical framework

3.1.1 Diffusion process fundamentals

This sub-section outlines the fundamental concepts of the diffusion process, providing the theoretical basis for understanding how noise is progressively added and removed in generative modeling.

3.1.2 Basic principles of diffusion models

Diffusion models constitute an innovative family of generative approaches that have risen to prominence due to their remarkable capacity for generating superior-quality samples across multiple data domains. These techniques are rooted in concepts from non-equilibrium statistical mechanics and function through iteratively transforming random noise into meaningful data via successive refinement operations. Compared to GANs and VAEs, diffusion models demonstrate superior training stability and generate outputs exhibiting enhanced diversity and quality, though they demand increased computational overhead during the generation phase.

Comprehending the operational mechanisms of diffusion models necessitates examining their core mathematical structure. Denoising Diffusion Probabilistic Models (DDPMs) (Ho et al. 2020) produce data via mechanisms that mirror physical diffusion phenomena, implementing dual interconnected processes: a forward corruption procedure and its inverse reconstruction counterpart, which together enable sophisticated data generation capabilities.

During the forward procedure implemented in DDPMs, a deterministic Markov process systematically introduces Gaussian perturbations to transform data into pure random noise. Beginning with $q(\mathrm{x}_0)$ representing the actual data distribution and $\mathrm{x}_0 \sim q(\mathrm{x}_0)$ denoting samples drawn from it, the forward procedure designated as $q(\mathrm{x}_{1:T}|\mathrm{x}_0)$ incrementally corrupts the data through noise injection, transitioning from x_0 to x_T:

$$q(\mathrm{x}_{1:T}|\mathrm{x}_0) := \prod_{t=1}^{T} q(\mathrm{x}_t|\mathrm{x}_{t-1}), \quad q(\mathrm{x}_t|\mathrm{x}_{t-1}) := \mathcal{N}(\mathrm{x}_t; \sqrt{\alpha_t}\mathrm{x}_{t-1}, \beta_t \mathrm{I}) \tag{1}$$

Here, β_t represents the noise variance introduced at each timestep, with the transformation $\alpha_t = 1 - \beta_t$. The cumulative product $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ is introduced according to Sohl-Dickstein et al. (2015). This formulation enables expressing the progression from initial data x_0 to intermediate state x_t through a Gaussian distribution:

$$q(\mathrm{x}_t|\mathrm{x}_0) = \mathcal{N}(\mathrm{x}_t; \sqrt{\bar{\alpha}_t}\mathrm{x}_0, (1 - \bar{\alpha}_t)\mathrm{I}) \tag{2}$$

This formulation characterizes how x_t relates to the original data x_0, showing that x_t can be represented as a weighted combination of the original data and noise, with $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathrm{I})$ denoting standard Gaussian noise:

$$\mathrm{x}_t = \sqrt{\bar{\alpha}_t}\mathrm{x}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon} \tag{3}$$

The inverse procedure seeks to progressively eliminate noise and recover the initial data through reversing the corruption applied during the forward procedure. This reconstruction is implemented as a Markov process where transitions from x_t to x_{t-1} utilize a parameterized conditional distribution $p_{\boldsymbol{\theta}}(\mathrm{x}_{t-1}|\mathrm{x}_t)$. The complete procedure is formulated as:

$$p_\theta(\mathrm{x}_{0:T}) := p(\mathrm{x}_T) \prod_{t=1}^{T} p_\theta(\mathrm{x}_{t-1}|\mathrm{x}_t), \quad p_\theta(\mathrm{x}_{t-1}|\mathrm{x}_t) := \mathcal{N}(\mathrm{x}_{t-1}; \mu_{\boldsymbol{\theta}}(\mathrm{x}_t, t), \Sigma_{\boldsymbol{\theta}}(\mathrm{x}_t, t)) \tag{4}$$

Here $p(\mathrm{x}_T)$ denotes the prior Gaussian distribution at timestep T, while $p_\theta(\mathrm{x}_{t-1}|\mathrm{x}_t)$ captures the learned transition distribution. The parameters $\mu_\theta(\mathrm{x}_t, t)$ and $\Sigma_\theta(\mathrm{x}_t, t)$ represent learned functions dependent on the current state x_t,

timestep t, and network parameters θ. During training, the goal involves minimizing the negative log-likelihood through the variational bound to match the target data distribution:

$$\mathbb{E}[-\log p_\theta(\mathrm{x}_0)] \leq \mathbb{E}_q\left[-\log \frac{p_\theta(\mathrm{x}_{0:T})}{q(\mathrm{x}_{1:T}|\mathrm{x}_0)}\right] = \mathbb{E}_q\left[-\log p(\mathrm{x}_T) - \sum_{t\geq 1}\log \frac{p_\theta(\mathrm{x}_{t-1}|\mathrm{x}_t)}{q(\mathrm{x}_t|\mathrm{x}_{t-1})}\right] =: L \tag{5}$$

This loss function breaks down the training task into KL divergence terms at individual timesteps, enabling step-by-step optimization of the reconstruction procedure. Through mathematical manipulation and leveraging the Gaussian structure, the optimization reduces to a streamlined mean squared error objective:

$$L_{\text{simple}}(\boldsymbol{\theta}) := \mathbb{E}_{t,\mathrm{x}_0,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta}}\left(\sqrt{\bar{\alpha}_t}\mathrm{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t\right)\right\|^2\right] \tag{6}$$

3.1.3 Video generation approaches

This sub-section categorizes the main strategies for adapting diffusion models to video, highlighting frame-level and sequence-level modeling methods along with mechanisms for ensuring temporal consistency.

Temporal modeling approaches: Video diffusion models must address the fundamental challenge of temporal consistency while maintaining spatial quality. The approaches can be broadly categorized into frame-level processing and sequence-level modeling.

Frame-level processing: *Grid-based video generation* Grid-based methodologies for video synthesis utilize 2D image diffusion architectures by transforming temporal dynamics into spatially-arranged representations. This dimensional reduction technique encodes the temporal evolution of videos into a structured grid format amenable to processing by conventional image-based generative models. As depicted in Fig. 3, the synthesis pipeline initiates with the construction of a key grid image a spatiotemporal representation comprising multiple temporally distinct frames arranged in a predefined spatial configuration that encapsulates the principal motion vectors and semantic transitions of the target video sequence.

The key grid image generation protocol, illustrated in Fig. 3, involves the extraction and spatial arrangement of four temporally equidistant frames from the video in chronological sequence. The resultant key grid image maintains a spatial resolution of 512×512 pixels, partitioned into four constituent frames, each with dimensions of 254×254 pixels. This representation exploits the inherent capacity of Variational Autoencoder (VAE) latent spaces to encode global spatial structures, thereby facilitating the preservation of temporal dynamics with equidistant intervals (Rombach et al. 2022). Empirical analysis confirms that this key grid representation effectively captures motion trajectories, enabling the model to accurately represent scene transitions and dynamic object motions.

To enhance temporal resolution and ensure inter-frame coherence, an autoregressive grid image interpolation mechanism is implemented. As illustrated in the figure, this process employs a hierarchical interpolation framework operating on masked grid images to recursively synthesize intermediate frames. The interpolation architecture comprises two specialized models operating in sequential order: (1) a 1-Step Interpolation model that performs coarse temporal interpolation by generating frames between temporally adjacent keyframes through strategic application of spatial masking, and (2) a 2-Step Interpolation model that executes fine-grained temporal refinement on the outputs of the 1-Step Interpolation to further enhance motion smoothness and temporal consistency.

Network inflation for video generation: Network inflation constitutes a sophisticated methodological paradigm for extending pre-trained text-to-image (T2I) diffusion models to the video generation domain. This approach centers on the systematic transformation of 2D architectures into spatio-temporal models through targeted modifications to attention mechanisms and convolutional layers while preserving the knowledge encapsulated in pre-trained weights. The underlying diffusion model typically employs a U-Net architecture (Ronneberger et al.

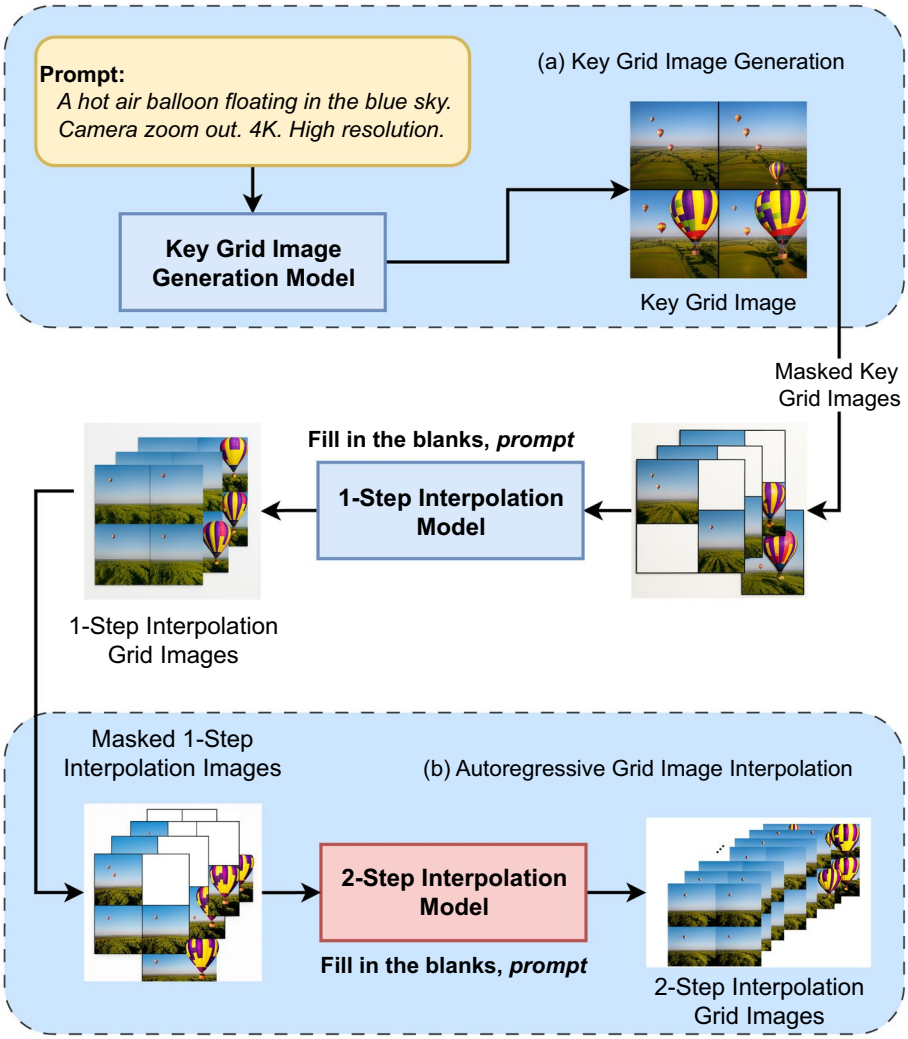


Fig. 3 Overview of the grid-based video synthesis pipeline. **a** Key Grid Image Generation: A text prompt is processed by the Key Grid Image Generation Model to create a 2×2 grid image containing four temporally equidistant keyframes. **b** Autoregressive Grid Image Interpolation: The process operates in two stages first, the 1-Step Interpolation Model fills in masked regions between keyframes to generate intermediate frames; subsequently, the 2-Step Interpolation Model performs further temporal refinement on masked 1-Step interpolation images to enhance motion smoothness and temporal consistency. This hierarchical approach enables the synthesis of coherent video sequences with preserved temporal dynamics. Figure adapted from (Lee et al. 2024)

2015) composed of stacked 2D convolutional residual blocks interspersed with transformer blocks, each comprising spatial self-attention layers, cross-attention mechanisms, and feed-forward networks (FFNs).

In the standard T2I diffusion framework, the spatial self-attention mechanism is formalized as:

$$\text{Attention}(\mathrm{Q}, \mathrm{K}, \mathrm{V}) = \text{Softmax}(\mathrm{QK}^T) \cdot \mathrm{V} \tag{7}$$

where the query, key, and value matrices are computed via:

$$\mathrm{Q} = \mathrm{W}^Q \mathrm{z}_{v_i}, \mathrm{K} = \mathrm{W}^K \mathrm{z}_{v_i}, \mathrm{V} = \mathrm{W}^V \mathrm{z}_{v_i} \tag{8}$$

with W^Q, W^K, and W^V representing learnable projection matrices that transform the latent representation z_{v_i} of video frame v_i into the respective attention components, and d denoting the output dimensionality of query and key features.

The inflation procedure extends the 2D Latent Diffusion Model (LDM) (Rombach et al. 2022) to the spatio-temporal domain through multiple strategic modifications. First, 2D convolutional layers are inflated to pseudo-3D convolutions, with 3×3 kernels being systematically replaced by $1 \times 3 \times 3$ kernels to incorporate temporal information while maintaining spatial processing capabilities. Second, temporal self-attention layers are appended to each transformer block to explicitly model inter-frame dependencies. Fig. 4 illustrates this inflation pipeline

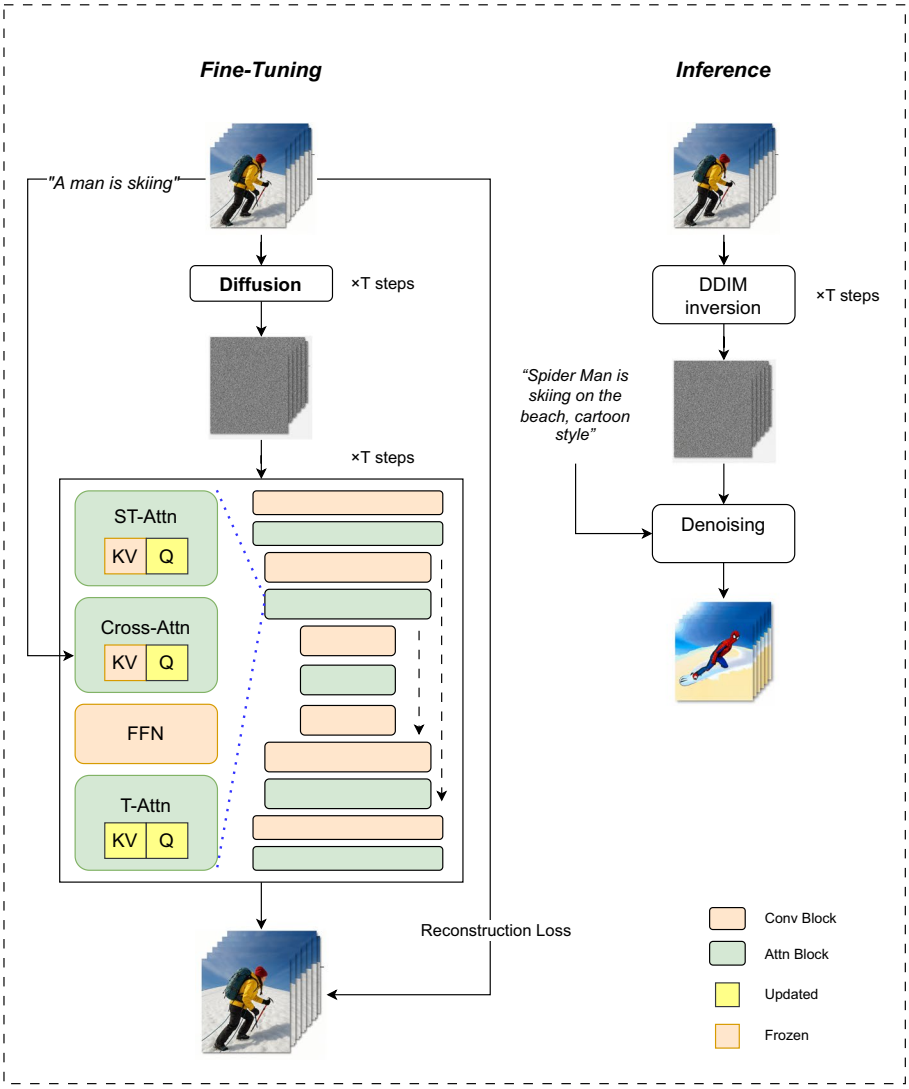


Fig. 4 Network inflation approach for video generation showing both fine-tuning and inference phases. The left side illustrates the fine-tuning process with selective parameter updates (yellow components), while the right side demonstrates the inference procedure with DDIM inversion and modified prompt for content editing. Figure adapted from (Wu et al. 2023)

for the Tune-A-Video framework, showing the selectively updated parameters during fine-tuning and the DDIM-inversion path used at inference time.

Sequence-level Modeling To achieve true spatio-temporal consistency, "video diffusion models need to share information across video frames" (Melnik et al. 2024). The canonical solution adds "a third temporal dimension to the denoising model" resulting in a noise prediction function:

$$\epsilon_\theta(\mathrm{x}_t, t, \mathrm{y}) : \mathbb{R}^{T\times H\times W\times C} \times \mathbb{R} \times \mathcal{Y} \to \mathbb{R}^{T\times H\times W\times C} \tag{9}$$

operating on the entire video volume $x_t \in \mathbb{R}^{T\times H\times W\times C}$ (with T frames of height H, width W, and C channels). Fig. 5 shows how the standard 2D U-Net is extended into this spatio-temporal form.

As illustrated in the figure, the 3D UNet architecture implements:

1. 3D convolutions through ResNet blocks:

$$\mathrm{Conv3D}(\boldsymbol{x})_{i,j,k,c'} = \sum_{l,m,n,c} \boldsymbol{w}_{l,m,n,c,c'} \cdot \boldsymbol{x}_{i+l,j+m,k+n,c} \tag{10}$$

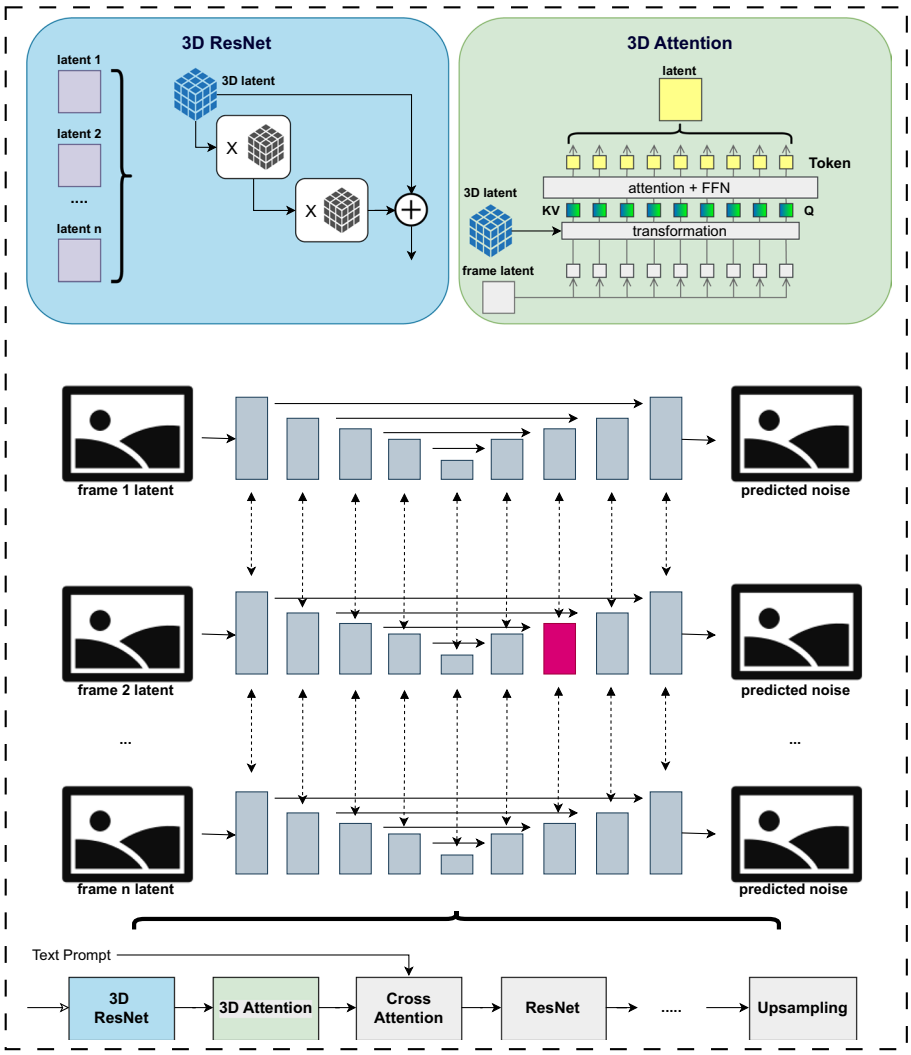


Fig. 5 Representation of 3D architecture of UNet. Figure adapted from (Melnik et al. 2024)

2. Self-attention across all frames simultaneously:

$$\text{Attention}(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) = \text{softmax}\left(\frac{\boldsymbol{QK^T}}{\sqrt{\boldsymbol{d_k}}}\right)\boldsymbol{V} \tag{11}$$

where $Q, K, V \in \mathbb{R}^{(T\cdot H\cdot W)\times d}$

Temporal Consistency Mechanisms Maintaining temporal consistency across frames represents a significant challenge in the generation and editing of videos utilizing diffusion models. Methodologies addressing this issue often leverage attention mechanisms, initialization protocols, conditioning frameworks, or specific architectural designs.

A primary category of methods focuses on the **modification or control of attention mechanisms**. For instance, cross-frame unified attention control enhances spatiotemporal consistency during video generation by applying the keys (K) and values (V) derived from the initial frame (K_0, V_0) to the self-attention computations of all subsequent frames, governed by the relation $\text{out} = \text{SelfAttention}(Q, K_0, V_0)$ (Chen et al. 2024).

A second category of approaches utilizes **consistent initialization and conditioning strategies**. The *Tune-A-Video* method, for example, exploits the observation that initializing the generation process for all frames from an identical noise map inherently promotes content consistency, necessitating only minimal temporal adjustments for motion synthesis (Wu et al. 2023).

A third strategy entails the implementation of **architectural constraints for alignment**, particularly pertinent in multi-modal contexts. Methodologies engineered to synchronize distinct modalities, such as audio and video, indirectly bolster temporal consistency within each individual modality.

3.1.4 Multi-modal integration theory

This sub-section reviews the theoretical principles and mechanisms by which multiple input modalities such as text, audio, and images are integrated within diffusion-based video generation frameworks.

Multi-Modal Diffusion: To illustrate the diversity of multi-modal integration in diffusion-based video generation, this section highlights several notable recent frameworks, categorized by the primary modalities involved.

Vision-Language (Text-to-Video): Text-conditioned video diffusion has witnessed rapid advancement. Foundational works such as *Imagen Video* (Ho et al. 2022) and *Make-A-Video* (Singer et al. 2022) first demonstrated high-quality text-to-video synthesis by extending image diffusion models. Alongside frameworks like *Tune-A-Video* (Wu et al. 2023) and *Stable Video Diffusion* (Blattmann et al. 2023), more recent contributions include *GridDiffusion*, which introduced a novel 2D grid methodology for text-to-video generation (Lee et al. 2024).

Audio-Visual (Audio/Text-to-Video): Multi-modal video generation frequently incorporates audio, either as the primary driver or in conjunction with text. *MM-Diffusion* exemplifies a joint audio-video diffusion model engineered to generate synchronized video and audio tracks concurrently (Ruan et al. 2023). This is achieved through coupled denoising networks for each modality, linked via cross-modal attention mechanisms to ensure alignment, such as correlating musical beats with visual motion.

Vision-to-Video (Image/Video-conditioned): Certain frameworks condition video diffusion processes on visual inputs, including initial frames, reference images, or draft video sequences. *UniAnimate*, for instance, conditions generation on a single input character image and a target pose sequence, producing a video where the character's appearance remains faithful to the input image while adhering to the specified motion trajectory (Wang et al. 2024b).

Cross-Modal Conditioning Mechanisms: Cross-modal conditioning mechanisms enable diffusion models to leverage information from multiple input modalities to guide the generation process. These mechanisms are essential for achieving coherent multi-modal video synthesis.

Attention-Based Cross-Modal Fusion: Cross-attention mechanisms allow the model to attend to relevant features from different modalities. In text-conditioned video generation, the visual features query textual embeddings to incorporate semantic guidance. The cross-attention operation can be formulated as:

$$\text{CrossAttention}(\boldsymbol{Q_v}, \boldsymbol{K_t}, \boldsymbol{V_t}) = \text{softmax}\left(\frac{\boldsymbol{Q_v}\boldsymbol{K_t^T}}{\sqrt{\boldsymbol{d_k}}}\right)\boldsymbol{V_t} \tag{12}$$

where Q_v represents visual queries, and K_t, V_t represent textual keys and values respectively.

Feature Concatenation and Projection: Another approach involves concatenating features from different modalities and using learned projections to align them in a common embedding space. This enables the model to process multi-modal information jointly while preserving modality-specific characteristics.

Adaptive Conditioning Strategies: Advanced conditioning mechanisms employ adaptive strategies that dynamically adjust the influence of different modalities based on the generation context. This allows for more nuanced control over the generation process and better handling of conflicting or incomplete modal information.

Pros, Cons, and When to Use Each Mechanism: The three cross-modal fusion mechanisms discussed above are not interchangeable; each exhibits distinct strengths and weaknesses that make it preferable for specific application profiles. **Attention-based cross-modal fusion** offers the highest semantic alignment quality because every spatio-temporal token can selectively attend to the condition, and it is the mechanism of choice when the condition is rich and mostly linguistic (text-to-video, dialogue-driven avatar generation). Its downside is quadratic memory cost in the number of tokens, which becomes prohibitive for long videos or dense pose conditions. **Feature concatenation with learned projection** scales linearly and is robust when modalities arrive at similar

granularity (e.g., image-plus-depth, where both are pixel-aligned); however, it enforces a symmetric treatment of modalities that can dilute the influence of the more informative signal and tends to underperform when one modality is much sparser than the other. **Adaptive conditioning** (gating, classifier-free guidance, and similar approaches) best handles *mixed-reliability* inputs—for example, reference image plus noisy pose—because the model can learn to down-weight the less reliable condition, but it requires more careful training regimes and additional hyper-parameters. In practice, recent high-performing systems combine the three: a concatenation layer for low-level alignment, attention for semantic control, and a gating scalar for robustness. Table 3 captures the analogous trade-offs at the *parameter-efficiency* level for fine-tuning these conditioning pathways.

3.2 Optimization and efficiency

The computational expense of diffusion models has prompted significant research into optimization strategies that maintain generation quality while improving efficiency. This section examines training optimization techniques and model compression approaches.

3.2.1 Training optimization

This sub-section details algorithmic and architectural strategies employed during training to improve the efficiency, stability, and convergence speed of diffusion models without sacrificing generation quality.

Training-phase algorithmic optimizations: Training-phase algorithmic optimizations in diffusion models focus on improving the foundational mathematical formulations and algorithms to reduce computational complexity, training time, memory usage, and energy consumption without sacrificing output quality. Unlike post-training techniques such as quantization and pruning, these optimizations target the core diffusion process itself. As illustrated in Fig. 6, these algorithmic optimizations can be broadly categorized into several key approaches including noise schedule optimization, advanced score matching techniques, data-dependent adaptive priors, and rectified flow methods.

Noise Schedule Optimization: Noise scheduling represents a critical component that governs how noise is added during the forward process and removed in the reverse process. The original DDPM model (Ho et al. 2020) employed a linear noise schedule, which while functional, required numerous timesteps and complex noise term calculations. Subsequent improvements introduced more sophisticated approaches such as the cosine noise schedule proposed in IDDPM (Nichol and Dhariwal 2021):

$$\beta_t = 1 - \frac{\cos\left(\frac{t}{T} + s \cdot \pi\right)^{\frac{1+s}{2}}}{\cos\left(s \cdot \frac{\pi}{2}\right)} \tag{13}$$

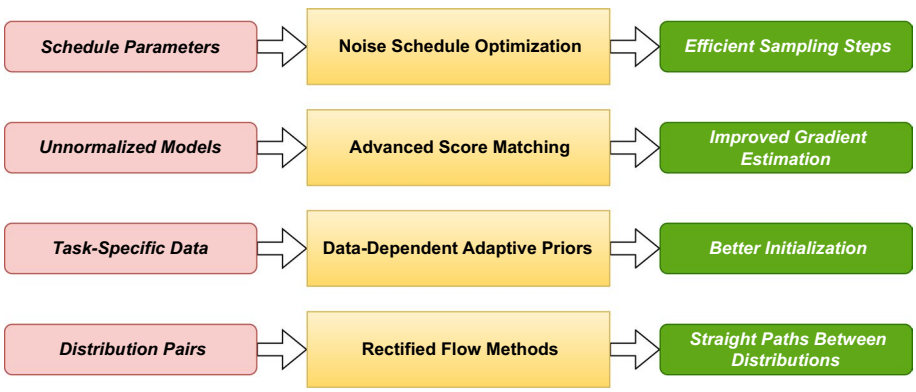


Fig. 6 Overview of training-phase algorithmic optimizations for diffusion models. The diagram illustrates the key optimization approaches: noise schedule optimization, advanced score matching, data-dependent adaptive priors, and rectified flow methods, along with their inputs and outcomes

Table 2 Comparison of different noise schedules for diffusion models

Schedule Type	Method	Key Characteristics	References
Fixed	Linear	Standard progression, uniform noise addition	Ho et al. (2020)
	Cosine	Smooth noise transition, better sampling efficiency	Nichol and Dhariwal (2021)
	Laplace	Resource concentration on medium noise intensities	Hang et al. (2024)
	Square-root	Stabilized training for text generation	Li et al. (2022)
Adaptive	DDIM	Non-Markovian processes, dynamic noise adjustment	Song et al. (2022)
	ResShift	Shorter Markov chains for super-resolution	Yue et al. (2023)
	Immiscible	Minimized distance between image-noise pairs	Li et al. (2024)
	Token-position	Dynamic adjustment per token position	Yuan et al. (2024)

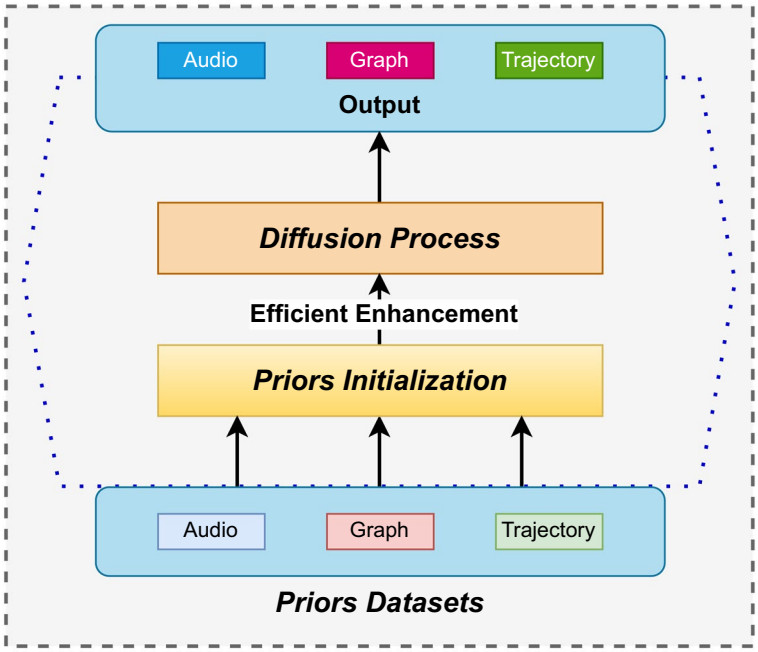


Fig. 7 Illustration of the data-dependent adaptive prior process. The diffusion model leverages prior datasets from multiple modalities (audio, graph, trajectory) for prior initialization, followed by efficient enhancement through the diffusion process to generate the final output. Figure adapted from (Shen et al. 2025)

where t is the current timestep, T is the total number of timesteps, and s is a small positive constant typically used to smooth the initial noise addition. Table 2 summarizes the fixed and adaptive noise schedules most widely used in recent diffusion models, together with their key characteristics.

Advanced Score Matching: Score matching represents another significant optimization avenue, directly estimating unnormalized models by minimizing the Fisher divergence between score functions. While traditional score matching avoids computing intractable partition functions, it faces computational challenges with high-dimensional data due to expensive Hessian matrix calculations. To address this, Sliced Score Matching (Song et al. 2020) projects score functions onto random directions, reducing computational complexity through dimensionality reduction.

Data-Dependent Adaptive Priors Traditional diffusion models typically assume standard Gaussian priors, which may not align with actual data distributions. By leveraging data-dependent priors, models achieve better initialization and faster convergence. PriorGrad (Lee et al. 2022) demonstrates this for speech synthesis, using adaptive priors derived from data statistics. Fig. 7 illustrates the overall workflow in which modality-specific prior datasets are used for initialization before the diffusion process refines the sample.

Rectified Flow Methods Rectified Flow methods (Liu 2022; Liu et al. 2022) provide another optimization pathway by learning straight transport paths between distributions, minimizing transport cost through direct routes rather than inefficient roundabout paths. This approach can be formalized as finding the optimal velocity field

$v_t(x)$ that minimizes the transport cost: The geometric intuition is shown in Fig. 8, contrasting the multi-step ODE trajectory with the straightened single-step transport enabled by Rectified Flow.

$$\min_{v} \mathbb{E}_{x_0 \sim \pi_0, t \sim \mathcal{U}[0,1]} \left[\| v_t(x_t) - \frac{x_1 - x_0}{1} \|^2 \right] \tag{14}$$

Efficient fine-tuning approaches: The computational expense of fine-tuning diffusion models has prompted the development of parameter-efficient methods that maintain model quality while reducing resource consumption. These approaches can be categorized into three primary paradigms: Low-Rank Adaptation (LoRA), Adapters, and ControlNet architectures. Table 3 summarizes the comparative trade-offs of these paradigms along six practical dimensions.

Low-Rank Adaptation (LoRA) LoRA (Hu et al. 2022) represents a parameter-efficient fine-tuning strategy that freezes pre-trained model weights while injecting trainable low-rank decomposition matrices into transformer layers. The mathematical foundation of LoRA can be formulated as:

$$\mathrm{W} = \mathrm{W}_0 + \Delta\mathrm{W}, \quad \text{where} \quad \Delta\mathrm{W} = \mathrm{BA} \tag{15}$$

Here, $\mathrm{W}_0 \in \mathbb{R}^{d \times k}$ represents the pre-trained weight matrix, while $\mathrm{B} \in \mathbb{R}^{d \times r}$ and $\mathrm{A} \in \mathbb{R}^{r \times k}$ are trainable low-rank matrices with rank $r \ll \min(d, k)$. Fig. 9 visualises this reparameterization.

Adapter Modules Adapters represent a different approach to efficient fine-tuning, introducing small network layers into pre-trained models while keeping original weights frozen. These lightweight modules are strategically positioned between normalization and feed-forward layers within transformer blocks. Fig. 10 contrasts the placement of adapter layers within a transformer block with the internal down-projection–nonlinearity–up-projection structure of the adapter itself.

ControlNet Architectures ControlNet (Zhang et al. 2023) represents a significant advancement in adding spatial conditioning to pre-trained diffusion models. The architecture introduces a novel approach where the original model parameters are locked (frozen) while adding a trainable copy of the encoder and middle block components. These copies are connected through "zero convolution" layers convolution layers with weights and biases initialized to zero that gradually develop parameters during training without disrupting model stability. Fig. 11 depicts this construction, where the trainable copy (right) is grafted onto the frozen Stable Diffusion U-Net (left) through zero-initialized convolutions.

The mathematical formulation is given by:

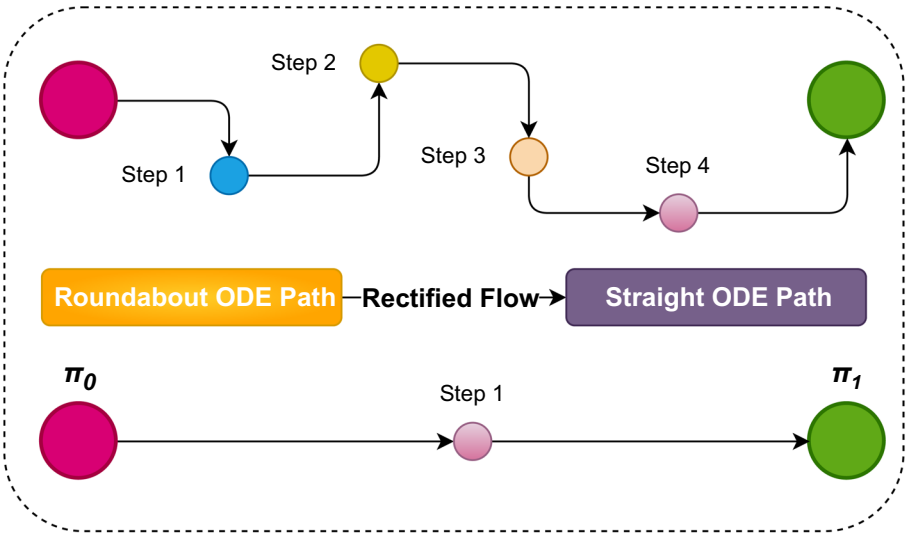


Fig. 8 Comparison between traditional ODE paths and Rectified Flow. The upper path shows a roundabout ODE trajectory with multiple intermediate steps between distributions π_0 and π_1, while the lower path demonstrates the straight ODE path enabled by Rectified Flow, allowing direct transport in a single step. Figure adapted from (Shen et al. 2025)

Article in Press

https://doi.org/10.1007/s10462-026-11699-z

Table 3 Comparison of Efficient Fine-Tuning Methods for Diffusion Models

Method	Parameter Efficiency	Compu-tation Overhead	Training Complexity	Control Precision	Multi-Modal Support	Key Applications
LoRA-based						
LCM-LoRA (Luo et al. 2023)	High	Low	Low	Medium	Text	Acceleration
Concept Sliders (Gandikota et al. 2024)	High	Low	Medium	High	Text	Attribute Control
LoRA-Composer (Yang et al. 2024)	High	Low	Medium	High	Text	Multi-Concept Integration
Adapter-based						
T2I-Adapter (Mou et al. 2024)	Medium	Medium	Medium	High	Multi-modal	Structural Guidance
IP-Adapter (Ye et al. 2023)	Medium	Medium	Medium	High	Image	Style Transfer
CTRL-Adapter (Lin et al. 2024)	Medium	Medium	Medium	High	Text	Attribute Control
ControlNet-based						
ControlNet (Zhang et al. 2023)	Low	High	High	Very High	Multi-modal	Spatial Conditioning
ControlNet++ (Li et al. 2024)	Low	High	High	Very High	Multi-modal	Alignment Optimization
ControlNet-XS (Zavadski et al. 2024)	Medium	Medium	Medium	High	Multi-modal	Efficient Guidance
ControlNeXt (Peng et al. 2025)	High	Low	Low	High	Multi-modal	Streamlined Architecture

Parameter Efficiency: High = trainable parameters < 5% of base model; Medium = 5–20%; Low = > 20%. *Computation Overhead*: Low = no inference-time cost beyond the frozen backbone; Medium = lightweight adapter or gating layer; High = an additional trainable encoder branch. *Training Complexity*: Low = single-stage with few hyper-parameters; Medium = multi-stage but standard optimizer; High = requires careful warm-up or zero-initialization. *Control Precision*: Medium = attribute-level control; High = spatial/structural control; Very High = dense pixel-level conditioning

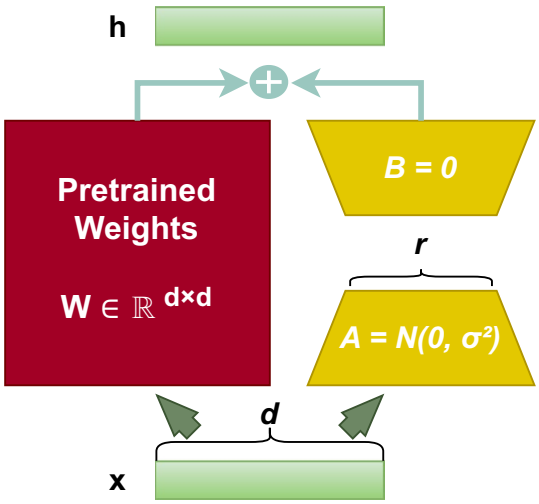


Fig. 9 LoRA reparameterization technique. The model keeps the pre-trained weights $W \in \mathbb{R}^{d \times d}$ frozen (red) while introducing trainable low-rank matrices A and B (yellow). The rank r is significantly smaller than the original dimension d, enabling efficient fine-tuning with reduced parameter count. Figure adapted from (Hu et al. 2022)

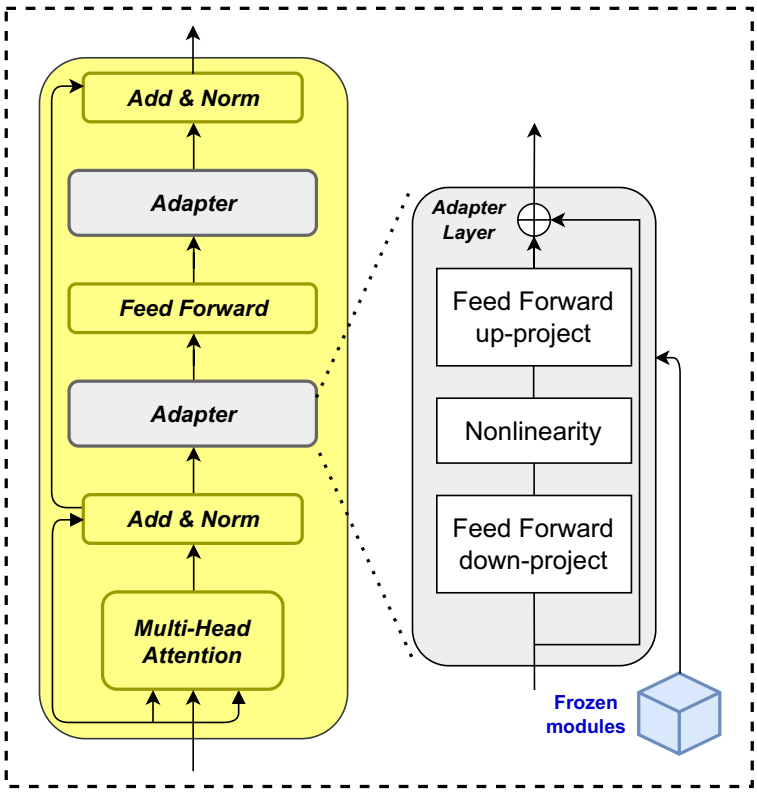


Fig. 10 Architecture comparison of adapter modules. Left: Traditional transformer-based adapter implementation with dedicated adapter layers positioned between multi-head attention, feed forward, and normalization layers (yellow modules). Right: Common adapter structure showing the internal components - feed forward down-projection, nonlinearity, and feed forward up-projection layers - with the adapter layer connecting to frozen pre-trained modules (grey region). Figure adapted from (Shen et al. 2025)

$$y_c = F(x;\Theta) + Z(F(x + Z(c;\Theta_{z1});\Theta_c);\Theta_{z2}) \tag{16}$$

where $F(x;\Theta)$ represents the original network block with locked parameters Θ, Θ_c are the trainable parameters of the copy, $Z(\cdot;\cdot)$ denotes the zero convolution operation, and c is the conditional input.

3.2.2 Model efficiency

This sub-section examines post-training techniques such as quantization and pruning that compress diffusion models for more efficient deployment while maintaining acceptable output quality.

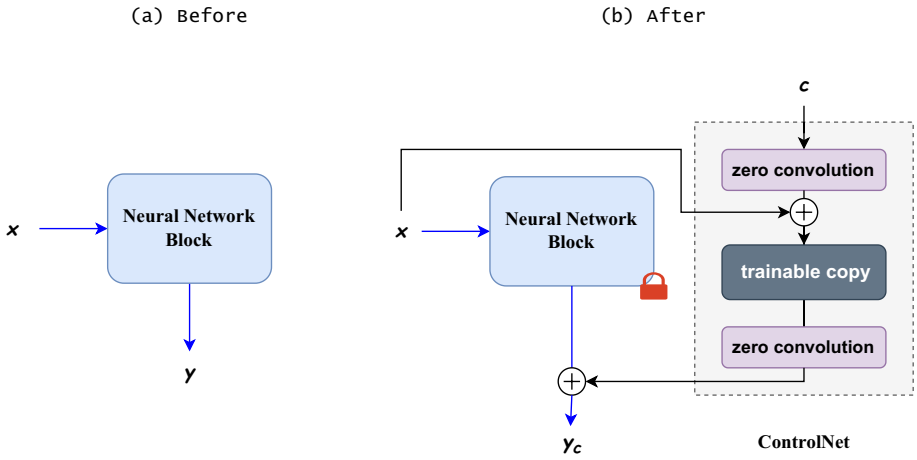


Fig. 11 ControlNet architecture applied to Stable Diffusion. Left: The original Stable Diffusion U-Net with encoder blocks, middle block, and decoder blocks. Right: The ControlNet structure adds trainable copies of encoder blocks and the middle block, connected with zero convolution layers. Image adapted from (Zhang et al. 2023)

Model Compression Techniques for Diffusion Models: Diffusion models, while effective for high-quality generation, face significant deployment challenges due to their computational demands. This section examines key compression techniques that aim to make these models more efficient while preserving generation quality.

Quantization: Quantization transforms the model's weights and activations from floating-point to low-bit representations, significantly reducing memory requirements and computational costs (Tian et al. 2024; Yahia et al. 2024).

Post-training Quantization (PTQ): PTQ methods convert pre-trained models to lower precision without requiring retraining or fine-tuning, making them particularly practical for deployment.

The standard quantization process maps floating-point values to their integer representations:

$$\boldsymbol{w_q} = \text{clamp}\left(\left\lfloor \frac{\boldsymbol{w}}{\boldsymbol{s}} \right\rceil + \boldsymbol{z}, 0, 2^{\boldsymbol{b}} - 1\right) \tag{17}$$

The dequantized approximation is then computed as:

$$\hat{\boldsymbol{w}} = \boldsymbol{s} \cdot (\boldsymbol{w_q} - \boldsymbol{z}) \approx \boldsymbol{w} \tag{18}$$

where w and $\hat{w}$ are the original and dequantized values, w_q is the quantized integer representation, s represents the scaling factor, z is the zero point, and b is the bit precision (Tian et al. 2024).

Quantization techniques specifically designed for diffusion models must address unique challenges not present in conventional neural networks. For video diffusion models, Tian et al. identify two critical challenges:

1. Temporal Feature Homogenization: Standard uniform quantizers can cause temporal features to collapse to similar values, severely degrading video quality. The proposed High Temporal Discriminability Quantization (HTDQ) utilizes logarithmic quantization to preserve the discriminability of temporal features (Tian et al. 2024).
2. Inter-Channel Variations: Activation values in temporal attention modules exhibit significant inter-channel disparities, resulting in low coverage of quantization levels. The Scattered Channel Range Integration (SCRI) method integrates activations across channels to improve quantization performance (Tian et al. 2024).***Pruning*** Pruning reduces model size by selectively removing less important parameters or components, leading to sparser models with reduced computational requirements.

Temporal Block Pruning: Ben Yahia et al. introduce an effective pruning strategy specifically for video diffusion models by identifying and removing less critical temporal blocks (Yahia et al. 2024). In their approach, each temporal block is assigned an importance value q_i, where $0 \leq q_i \leq 1$, indicating its contribution to the model's performance. Only the n blocks with the highest importance values are retained during inference.

The output of a temporal block in the original architecture is formulated as:

$$\alpha \boldsymbol{x_s} + (1-\alpha)\boldsymbol{x_t} = \boldsymbol{x_s} + (1-\alpha)\boldsymbol{r_t} \tag{19}$$

where x_s and x_t are outputs of spatial and temporal blocks, and α is a learnable weight scalar (Yahia et al. 2024).

This method has shown remarkable effectiveness, allowing for the removal of up to 70% of temporal blocks with minimal impact on generation quality. Further pruning leads to significant degradation, suggesting a threshold beyond which model capacity becomes insufficient (Yahia et al. 2024).

Channel Funneling: Another pruning approach, channel funneling, reduces the width of neural network layers by introducing funnel matrices between consecutive layers. For two linear layers with weights $W_1 \in \mathbb{R}^{c_{inner} \times c_{in}}$ and $W_2 \in \mathbb{R}^{c_{out} \times c_{inner}}$, funnel matrices $F_1 \in \mathbb{R}^{c' \times c_{inner}}$ and $F_2 \in \mathbb{R}^{c_{inner} \times c'}$ are inserted, where $c' < c_{inner}$ (Yahia et al. 2024).

The network computation becomes:

$$\boldsymbol{y'} = \boldsymbol{W_2 F_2 \sigma(F_1 W_1 x)} \tag{20}$$

During inference, these funnel matrices can be merged with their associated weight matrices, resulting in a more compact model with reduced inner dimensions (Yahia et al. 2024).

Combined Approaches and Efficiency Gains The combination of quantization and pruning techniques can lead to multiplicative efficiency gains. Ben Yahia et al. demonstrate that their MobileVD model, incorporating temporal block pruning, channel funneling, and adversarial finetuning, achieves a 523× reduction in computational cost (from 1817.2 to 4.34 TFLOPs) with only a slight quality drop (FVD 149 vs. 171) compared to the original Stable Video Diffusion model (Yahia et al. 2024).

3.3 Conditional generation applications

The ability to control and guide the generation process represents one of the key advantages of diffusion models. This section examines how conditioning mechanisms enable precise control over content and style in video generation applications.

3.3.1 Content control

Style Transfer and Control: Style transfer in video diffusion aims to modify the appearance or style of a source video while preserving its content and temporal coherence. In diffusion terms, we seek a conditional generative process $x_{t-1} = f_\theta(x_t, t, c)$ that steers a noised video x_t towards a new style (condition c) without disrupting structure. Formally, many methods still optimize a denoising objective:

$$\mathbb{E}_{t,\epsilon}[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon_\theta}(\boldsymbol{x_t}, \boldsymbol{t}, \boldsymbol{c})\|^2] \tag{21}$$

for some UNet ϵ_θ, while introducing new conditioning c and losses to enforce content consistency.

Text-Driven Stylization: Style-A-Video (Huang et al. 2024) introduces a zero-shot text-based video stylization method using a latent diffusion model. The method modifies the guidance term:

$$\hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}} = \boldsymbol{\epsilon_\theta}(\boldsymbol{x_t}, \boldsymbol{t}, \boldsymbol{c}_{\text{style}}) + \boldsymbol{w}\left(\boldsymbol{\epsilon_\theta}(\boldsymbol{x_t}, \boldsymbol{t}, \boldsymbol{c}_{\text{style}}) - \boldsymbol{\epsilon_\theta}(\boldsymbol{x_t}, \boldsymbol{t}, \boldsymbol{c}_\emptyset)\right) \tag{22}$$

where w balances style injection against structural preservation. Temporal consistency is maintained through optimized sampling and a temporal consistency module.

One-Shot Personalization: Tune-A-Video (Wu et al. 2023) adapts a pretrained image diffusion model to video using spatio-temporal attention. DDIM inversion produces a structured noise prior x_T, and generation is performed via:

$$\hat{\boldsymbol{x}}_0 = \text{DDIM-Solve}(\boldsymbol{x_T}, \boldsymbol{c}_{\text{new}}) \tag{23}$$

The updated denoising model leverages inverted trajectories to control motion while altering appearance via c_{new}.

Structure-Guided Synthesis: Esser et al. (Esser et al. 2023) introduce a dual-conditioning strategy where structure s (e.g., depth maps) and content c_{cont} are input to the model via:

$$\epsilon_{\theta}([\boldsymbol{x}_t \oplus \boldsymbol{s}], \boldsymbol{t}, \boldsymbol{c}_{\text{cont}}) \tag{24}$$

with structure s concatenated channel-wise. The model employs a learned guidance parameter t_s to vary spatial detail during training and inference.

Pose and Trajectory Control: VividPose (Wang et al. 2024a), MotionCtrl (Wang et al. 2024b), and HumanDiT (Gan et al. 2025) provide conditioning via keypoint sequences or 6-DoF trajectories. Pose is encoded to latent maps P_t, which are injected into ϵ_θ using gated attention or spatial modulation:

$$\boldsymbol{H}'^{(l)} = \text{Attn}(\boldsymbol{H}^{(l)}, \boldsymbol{K}_{\text{pose}}, \boldsymbol{V}_{\text{pose}}) \cdot \boldsymbol{\sigma} + \boldsymbol{H}^{(l)} \tag{25}$$

ControlNet-Based Composition Modular injection of control signals $F_\phi(c)$ occurs via a frozen base model and learned adapter (Zhang et al. 2023):

$$\epsilon_{\theta}(\boldsymbol{x}_t, \boldsymbol{t}, \boldsymbol{c}) = \epsilon_{\theta}([\boldsymbol{x}_t \oplus \boldsymbol{F}_{\phi}(\boldsymbol{c})], \boldsymbol{t}) \tag{26}$$

Here, F_ϕ processes modality-specific inputs such as masks, edges, or depth.

Action-Conditioned Generation: In action-conditioned diffusion, the generative process is directed by explicit signals describing dynamics (Alonso et al. 2024). The denoising model takes form:

$$\epsilon_{\theta}(\boldsymbol{x}_t, \boldsymbol{t}, \boldsymbol{c}_{\text{action}}) \tag{27}$$

where c_{action} may consist of trajectories, action vectors, or audio.

Transition-Based Generation: MAVIN (Zhang et al. 2024) performs video infilling using boundary frame guidance and Gaussian Filter Mixing. During training, temporally consecutive noising is used. At inference, frames are initialized:

$$\hat{\boldsymbol{x}}_t = \boldsymbol{G}(\boldsymbol{x}_0^{\text{start}}, \boldsymbol{x}_T^{\text{end}}) + \mathcal{N}(0, \boldsymbol{\sigma}^2 \boldsymbol{I}) \tag{28}$$

where G is a Gaussian interpolation.

Action-Driven World Models: AVID (Rigter et al. 2024) introduces an adapter $\phi(a)$ that modulates UNet layers:

$$\boldsymbol{f}_{\text{mod}}^{(l)} = \phi(\boldsymbol{a}) \odot \boldsymbol{f}^{(l)} \tag{29}$$

allowing fine-grained control without modifying base weights.

Trajectory Conditioning: Motion Prompting (Geng et al. 2025) uses trajectories $\mathrm{u}_i(t)$ encoded as images M_t processed via a ControlNet:

$$\epsilon_{\theta}(\boldsymbol{x}_t, \boldsymbol{t}, \boldsymbol{c}_{\text{text}}, \boldsymbol{c}_{\text{motion}}) = \epsilon_{\theta}([\boldsymbol{x}_t \oplus \boldsymbol{F}_{\phi}(\boldsymbol{M})], \boldsymbol{t}, \boldsymbol{c}_{\text{text}}) \tag{30}$$

Disentangled Motion Control: MotionCtrl (Wang et al. 2024b) separates camera and object motion using:

$$\epsilon_{\theta}(\boldsymbol{x}_t, \boldsymbol{t}, \boldsymbol{c}_{\text{cam}}, \boldsymbol{c}_{\text{obj}}) \tag{31}$$

Temporal tokens represent c_{cam}, while object maps inject c_{obj} into spatial layers.

Audio-Driven Generation: DiffTED (Hogue et al. 2024) and CyberHost (Lin et al. 2025) use audio-derived features c_{audio} and region priors c_{codebook}:

$$\boldsymbol{L} = \mathbb{E}[\|\epsilon - \epsilon_{\theta}(\boldsymbol{x}_t, \boldsymbol{t}, \boldsymbol{c}_{\text{audio}}, \boldsymbol{c}_{\text{ref}})\|^2] + \boldsymbol{\lambda}_{\text{face}} \boldsymbol{L}_{\text{face}} + \boldsymbol{\lambda}_{\text{hand}} \boldsymbol{L}_{\text{hand}} \tag{32}$$

The attention mechanism targets local regions using VQ embeddings.

These models enable fine-grained multimodal control over motion, leveraging trajectories, poses, and audio signals to govern spatiotemporal generation.

3.4 Connections to large language models and broader multimodal learning

Video diffusion does not evolve in isolation. The design choices surveyed above—cross-modal attention, feature-space fusion, and adaptive conditioning—mirror and increasingly converge with developments in the wider LLM-driven multimodal research landscape, and a complete picture requires situating video diffusion within that landscape.

Three lines of work are directly relevant. First, **unified multimodal LLMs** such as NExT-GPT (Wu et al. 2024) and Show-o (Xie et al. 2025) demonstrate that a single transformer can accept and generate across text, image, video, and audio, blurring the traditional boundary between multimodal *understanders* and *generators*. The survey of Wang et al. (2025) analyzes this convergence in depth and identifies autoregressive-versus-diffusion and dense-versus-MoE as the central design axes. Second, the **fusion of LLMs with diffusion models** has emerged as a distinct research frontier: Benjdira et al. (2026) catalog techniques for latent-space alignment, LLM-as-planner prompt engineering, and hybrid architectures, and identify computational cost, modality misalignment, and data scarcity as the open problems shared with video diffusion. Third, **cross-domain multi-attention and tag-enriched representations**—originally developed for recommendation systems and weakly supervised segmentation (Wu et al. 2026, 2025a, b)—show that LLM-generated semantic tags can enrich low-label visual representations, a technique that transfers directly to pose-label sparsity and long-video annotation bottlenecks in the human-centric video setting.

These connections matter for three reasons relevant to the remainder of this survey. (i) Agent-based video pipelines (Sect. 2.1) inherit their coordination primitives from multi-agent LLM systems, making the LLM a substrate rather than merely a text encoder. (ii) Evaluation frameworks developed for multimodal LLMs—particularly those testing cross-modal grounding—inform the evaluation guidelines we propose in Sect. 6. (iii) The convergence toward unified models suggests that future human-centric video generation may be conditioned on richer instruction streams than the text-plus-pose paradigm currently dominant, a direction we return to in Sect. 7.

4 Human-centric diffusion models for video generation

Human-centric video generation has emerged as one of the most challenging yet important applications of generative AI, with diffusion models leading recent advancements in this domain. Unlike general video generation, human-centric approaches must address unique requirements including anatomical accuracy, temporal consistency, and realistic human behavior modeling (Feng et al. 2023; Lei et al. 2024). The field has seen rapid progression from initial single-person static pose generation to dynamic, interactive, and physically plausible human movement synthesis. However, as we will explore, achieving high-quality human-centric video generation requires addressing fundamental technical challenges while developing sophisticated methodological approaches that balance competing objectives.

4.1 Core technical challenges

Human-centric video generation faces four fundamental technical challenges that distinguish it from general video synthesis. These challenges stem from the complex nature of human appearance, motion, and social behavior, requiring specialized approaches that go beyond standard diffusion model architectures. Understanding these core challenges is essential for appreciating the methodological innovations discussed in Sect. 4.3.

4.1.1 Anatomical accuracy and physical constraints

A primary limitation of current human motion generation models is the frequent production of movements that, while visually plausible at first glance, violate fundamental laws of physics or human biomechanics (Tashakori et al. 2025; Yuan et al. 2023; Kang et al. 2025). Common artifacts include characters appearing to float above the ground, feet sliding unnaturally during locomotion, limbs penetrating objects or the ground, impossible joint rotations, and a general lack of convincing weight and dynamics.

Standard video or motion diffusion models typically lack innate knowledge of physics or the complex constraints governing human movement (Yuan et al. 2023; Ren et al. 2025). While they learn from motion capture data, this data may not perfectly capture all physical nuances, may contain artifacts itself, or may not cover the full spectrum of humanly possible (and impossible) motions (Janson et al. 2024). This gap between learned statistical patterns and physical reality creates a fundamental challenge: how to ensure that generated human motion adheres to both biomechanical constraints and perceptual expectations of naturalness.

The problem is compounded by the "uncanny valley" effect of physics enforcement. Human perception of naturalness is incredibly sensitive (Tevet et al. 2022), and motions that strictly adhere to a simplified or slightly inaccurate physical model might paradoxically appear *less* natural than motions generated by purely data-driven models. This creates a delicate balance between enforcing explicit physical rules and capturing the nuanced, complex, and sometimes seemingly non-optimal patterns of biological movement learned implicitly from data.

4.1.2 Identity preservation vs. motion dynamics trade-off

Achieving strong identity preservation while simultaneously generating highly dynamic and expressive motion presents an inherent tension in human-centric video generation. Techniques designed to rigidly enforce appearance consistency across frames can inadvertently hinder the model's ability to synthesize natural deformations, particularly in facial expressions or subtle body language (Xu et al. 2024; Zhang et al. 2025).

Real human motion, especially expressive facial movement, involves complex, non-rigid changes in appearance. If a model is overly constrained to match a single static reference image, it may struggle to generate these dynamic changes realistically, potentially resulting in a "mask-like" effect or unnaturally stiff movements. Conversely, models that allow greater flexibility in appearance features to achieve more dynamic and expressive motion might fail to consistently maintain the core visual identity of the person, leading to identity drift over the course of the video.

This fundamental trade-off stems from the conflicting objectives of maintaining static identity cues while enabling dynamic expression. The challenge lies in disentangling different aspects of human representation: the core, relatively static identity features; the underlying structural information such as facial landmarks and skeletal pose; and the dynamic, time-varying expression and motion components. Current approaches must navigate between preserving recognizable identity markers and allowing sufficient flexibility for natural human expression and movement.

4.1.3 Temporal consistency and motion style control

While current research focuses heavily on generating specific actions (e.g., "walking," "sitting") and preserving identity, the ability to control the *style* or *manner* in which an action is performed remains relatively underexplored. Generating a "sad walk" versus an "energetic walk," or controlling the specific style of a dance based on textual descriptions represents a more nuanced level of control that is currently difficult to achieve reliably.

The challenges lie in defining abstract style concepts in a way that models can understand and condition upon, and effectively disentangling these stylistic elements from the core action semantics and the character's identity. Unlike discrete action labels or pose sequences, style represents a continuous, subjective quality that varies along multiple dimensions such as energy level, emotional expression, cultural context, and individual mannerisms.

Table 4 contrasts the style-control capabilities of current methods along the dimensions of style input type, emotional control, and cultural variation, making explicit the gap left by methods that primarily target action accuracy.

Additionally, maintaining temporal consistency while allowing for style variation poses significant technical challenges. Models must ensure that generated motions remain coherent across time while exhibiting the desired stylistic qualities consistently throughout the sequence. This requires sophisticated temporal modeling that can distinguish between intentional style variations and unwanted temporal artifacts or inconsistencies.

Current approaches often treat style control as a secondary consideration, focusing primarily on action accuracy and identity preservation. This gap represents a significant limitation for applications requiring expressive human animation, such as entertainment, virtual avatars, and storytelling applications.

4.1.4 Multi-person scene complexity

Generating realistic interactions between multiple humans introduces exponentially greater complexity compared to single-person scenarios. Multi-person scenes require models to understand and generate coordinated behaviors, social dynamics, spatial relationships, and the complex interplay between individual actions and group dynamics.

The technical challenges encompass several dimensions. First, spatial coordination requires ensuring that multiple characters occupy space realistically without interpenetration while maintaining appropriate social distances and interaction zones. Second, temporal synchronization demands that individual character motions are coordinated appropriately for example, ensuring that conversational gestures, eye gaze, and speaking patterns align naturally between participants.

Social behavior modeling presents perhaps the most complex challenge, as human interactions follow subtle but important social conventions that vary by context, relationship, and cultural setting. Models must capture the nuanced dynamics of turn-taking in conversation, appropriate physical contact, coordinated group movements, and the complex feedback loops that characterize real human social interaction.

Furthermore, the computational complexity scales dramatically with the number of individuals in the scene. Each additional person not only increases the dimensional space of possible configurations but also introduces new interaction possibilities that must be modeled consistently. This scaling challenge becomes particularly acute for crowd simulation scenarios where dozens or hundreds of individuals must exhibit both realistic individual behaviors and emergent group dynamics.

The lack of comprehensive multi-person datasets compounds these challenges, as collecting and annotating realistic multi-person interaction data is significantly more expensive and technically demanding than single-person motion capture. Existing datasets often focus on specific interaction types or contain limited variation in social dynamics, constraining the generalization capabilities of trained models.

4.2 Unifying principles: why current methods fail

The four challenges above—anatomy/physics, identity–motion trade-off, temporal style control, multi-person complexity—are usually treated as independent research problems. Looking across the methodological landscape

Table 4 Motion Style Control Capabilities in Current Methods

Method	Style Input Type	Emotional Control	Cultural Variation	Style Control Limitations
Follow Your Pose (Ma et al. 2024)	Text descriptions	Limited	×	Action-focused, minimal style nuance
DreaMoving (Feng et al. 2023)	Text + pose	Basic	×	Style secondary to pose control
MimicMotion (Zhang et al. 2024)	Pose sequence	×	×	No explicit style conditioning
Text2Performer (Jiang et al. 2023)	Text descriptions	Basic	×	Limited style vocabulary in training
Human Motion Diffusion (Tevet et al. 2022)	Text descriptions	Basic	×	Style-action entanglement issues

Table 5 Comparison of physics integration approaches

Method	Physics Model Type	Integration Method	Constraints Enforced	Inference Cost
PhysDiff (Yuan et al. 2023)	External physics simulator	Projection during denoising	Contact forces, dynamics	High (simulation required)
BioVAE (Kang et al. 2025)	Biomechanical residual model	Residual VAE refinement	Biomechanical plausibility, joint limits	Low–medium (refinement)
FlexMotion (Tashakori et al. 2025)	Learned physics embedding	Latent space constraints	Joint limits, contact forces	Low (physics-free inference)
InterDiff (Xu et al. 2023)	Object interaction model	Two-stage correction	Collision avoidance, stability	Medium (correction step)
Human Motion Diffusion (Tevet et al. 2022)	Geometric constraints	Loss function regularization	Joint angle limits	Low (loss-based only)
Anatomical Control (Konz et al. 2024)	Anatomical segmentation	Mask conditioning	Anatomical structure	Low (mask-based guidance)

Table 6 Identity preservation technical approaches

Method	Identity Encoding	Disenta-nglement Strategy	Multi-Modal	Key Technical Innovation
MagicAnimate (Xu et al. 2024)	Dedicated appearance encoder	Parallel processing paths	×	Separate temporal and appearance encoding
Magic Mirror (Zhang et al. 2025)	Dual-branch facial extractor	Identity-structure separation	×	Transformer adapter integration
EchoVideo (Wei et al. 2025)	Identity-Text Fusion Module	Semantic representation	✓	High-level identity semantics
MotionCharacter (Fang et al. 2024)	ID-preserving modules	Region-aware processing	×	Selective attribute modification
Consistent Generation (Cao et al. 2024)	Reference feature queries	Causal feature interaction	×	Spatially-conditioned inpainting
DisPose (Li et al. 2025)	Motion field generation	Sparse-dense pose separation	×	Body shape adaptation
MagicPose (Chang et al. 2024)	Two-stage disentanglement	Motion-appearance separation	×	Expression and pose control
ID-Animator (He et al. 2024)	Zero-shot customization	Feature space manipulation	×	Training-free identity adaptation

(Tables 5, 6, 4), three deeper principles emerge that explain why existing methods fail in predictable, rather than ad-hoc, ways.

Principle 1: The Reference-Flexibility Trade-off: Any mechanism that rigidly anchors the generation to a reference signal (a face image, a pose skeleton, a physics simulator) improves *what it anchors* but harms *what it does not*. MagicAnimate (Xu et al. 2024) preserves identity but produces mask-like faces; PhysDiff (Yuan et al. 2023) enforces physics but produces stiff gait; strict pose guidance preserves trajectory but penalises different body shapes. Future designs should treat every reference signal as a *probabilistic prior* with a learned noise model, not as a hard constraint.

Principle 2: The Physics-Perception Asymmetry: Human perceptual sensitivity is non-uniform across motion attributes—we tolerate minor physical violations (slight foot slide) but not minor perceptual violations (unnatural facial micro-expression). Methods that invest compute uniformly in physics (InterDiff Xu et al. 2023) or uniformly in appearance (EchoVideo Wei et al. 2025) miss this asymmetry. The unifying remedy is *perceptually-weighted* constraint enforcement, spending simulation budget on contact and balance, and learned data budget on expression and gaze.

Principle 3: Disentanglement Granularity: Most identity/motion disentanglement methods operate at a single granularity: MagicAnimate (Xu et al. 2024) separates appearance and motion; Magic Mirror (Zhang et al. 2025) separates identity and structure; MotionCharacter (Fang et al. 2024) separates ID-critical and attribute features. Each succeeds only when the target scenario happens to match its chosen granularity. The unifying insight is that disentanglement should be *hierarchical*—from static identity at the top level, through structural pose and

anatomical configuration in the middle, down to dynamic micro-expression and style at the bottom—with adaptive weighting that depends on the query.

These three principles provide explanatory leverage: an "uncanny valley" artefact, for instance, is almost always either a Principle-1 rigidity effect, a Principle-2 asymmetry failure, or a Principle-3 granularity mismatch. They also suggest that the next generation of human-centric video diffusion models should be organised around *adaptive, hierarchical, perceptually-weighted* conditioning rather than around larger and stricter constraint sets.

4.3 Methodological approaches

To address the core challenges identified in Sect. 4.1, researchers have developed sophisticated methodological approaches that extend standard diffusion models with specialized architectures, training procedures, and conditioning mechanisms. These approaches can be categorized into four main methodological families, each targeting specific aspects of human-centric video generation while often addressing multiple challenges simultaneously. Table 8 provides a consolidated overview of representative methods across these four families, showing which core challenge each method primarily addresses.

4.3.1 Physics-informed generation methods

Addressing the anatomical accuracy and physical constraint violations discussed in Sect. 4.1, physics-informed diffusion methods integrate explicit physical models and biomechanical constraints into the generation process. These approaches aim to eliminate common artifacts such as floating, foot sliding, and impossible joint configurations while maintaining the generative flexibility of diffusion models.

PhysDiff: (Yuan et al. 2023) introduces a physics-based motion projection module that uses simulation to enforce physical constraints during the diffusion process. The key innovation lies in integrating a physics simulator directly into the denoising steps, allowing the model to correct physically implausible predictions in real-time. This approach demonstrates how external physics engines can be incorporated into diffusion training and inference without requiring the model to learn physical laws from data alone.

BioVAE: (Kang et al. 2025) takes a biomechanics-guided residual approach: a biomechanical variational autoencoder learns a residual correction that is applied on top of a motion diffusion backbone, improving the physical and biomechanical plausibility of generated motion while remaining generalizable across motion generation, pose estimation, and motion editing. This refinement-based design enforces biomechanical principles without requiring a physics simulator at inference, addressing the gap between high-level motion intentions and low-level physical plausibility.

FlexMotion: (Tashakori et al. 2025) incorporates joint locations, contact forces, and muscle activations within a latent diffusion framework to ensure physical plausibility without requiring a full physics simulator during inference. This approach addresses the computational overhead associated with physics simulation by learning to embed physical constraints within the latent representation itself. The method demonstrates how physics-aware training can produce models that generate physically plausible motion without explicit physics computation at inference time.

InterDiff: (Xu et al. 2023) extends physics-informed generation to human-object interactions, implementing a two-stage approach: interaction diffusion to model future interactions, followed by physics-informed correction to ensure physical validity. This framework demonstrates how physics constraints can be applied beyond single-person motion to complex interactive scenarios. Table 5 summarises these physics-integration approaches along four dimensions (physics-model type, integration method, constraints enforced, and inference cost), making explicit the trade-off between constraint strength and computational overhead.

These physics-informed approaches collectively address the uncanny valley problem identified in Sect. 4.1 by providing principled methods for balancing data-driven learning with physical realism. However, the integration

of physics constraints requires careful tuning to avoid overly rigid motions that, while technically correct, may appear unnatural to human observers.

4.3.2 Identity-preserving architectures

To address the identity preservation versus motion dynamics trade-off discussed in Sect. 4.1, several architectural innovations have been developed that enable consistent identity maintenance while allowing natural motion expression. These approaches typically employ specialized encoding mechanisms, disentanglement strategies, and multi-modal fusion techniques.

Magic Animate: (Xu et al. 2024) employs a dedicated video diffusion model for temporal encoding combined with a novel appearance encoder designed to capture fine-grained identity details from reference images. The architecture separates identity encoding from motion generation through parallel processing pathways, allowing independent optimization of identity consistency and motion quality. This separation addresses the fundamental trade-off by preventing identity constraints from overly restricting motion generation.

Magic Mirror: (Zhang et al. 2025) introduces a dual-branch facial feature extractor that separately processes identity information and structural facial information. This architectural innovation represents a key advancement in disentangling static identity cues from dynamic structural changes. The method incorporates a lightweight adapter within a Video Diffusion Transformer architecture, demonstrating how identity preservation can be integrated into modern transformer-based video generation frameworks.

Echo Video: (Wei et al. 2025) addresses the "copy-paste" artifacts common in identity-preserving methods through its Identity Image-Text Fusion Module (IITF). This approach integrates high-level semantic features from text with identity information, "capturing clean facial identity representations while discarding occlusions, poses, and lighting variations." The multi-modal approach represents a shift from purely visual identity preservation to semantic identity representation, potentially offering greater robustness to variation in reference images.

Motion Character: (Fang et al. 2024) introduces specialized ID-preserving modules combined with ID-consistency and region-aware loss mechanisms. The framework allows flexible attribute modifications while maintaining core identity features, demonstrating how architectural design can enable controlled modification of non-essential appearance aspects while preserving identity-critical features. A side-by-side comparison of these identity-preservation strategies—their encoding mechanism, disentanglement strategy, multi-modality, and key innovation—is given in Table 6.

Spatially-Conditioned Approaches: Recent work by Cao et al. (Cao et al. 2024) frames identity preservation as a spatially-conditioned inpainting problem, where target images are generated to maintain appearance consistency with reference images. Their causal feature interaction framework ensures that "reference features can only query from themselves, while target features can query appearance information from both the reference and the target," providing a principled approach to information flow in identity-preserving generation.

DisPose (Li et al. 2025) addresses identity preservation through disentangled pose guidance, recognizing that strict dense guidance can impair video quality when body shapes differ significantly between reference and target. Their approach generates dense motion fields from sparse skeleton poses, allowing flexible adaptation to different body types while maintaining identity consistency.

These identity-preserving architectures collectively demonstrate various strategies for navigating the identity-motion trade-off, ranging from parallel processing architectures to sophisticated disentanglement mechanisms and multi-modal fusion approaches.

4.3.3 Multi-person interaction frameworks

Addressing the multi-person scene complexity discussed in Sect. 4.1, several specialized frameworks have been developed to handle the coordination, social dynamics, and spatial relationships required for realistic multi-person video generation. These approaches range from dyadic interaction modeling to large-scale crowd simulation.

Social Behavior Modeling: DIM (Dyadic Interaction Modeling) (Tran et al. 2024) recognizes that "human-human communication is like a delicate dance where listeners and speakers concurrently interact to maintain conversational dynamics." The framework employs masking and contrastive learning to capture dyadic context, representing a significant advancement in modeling the subtle coordination patterns that characterize natural human conversation. This approach demonstrates how machine learning techniques can be adapted to capture the temporal synchronization and turn-taking behaviors essential to realistic social interaction.

Multi-Identity Generation: Wang et al. (Wang et al. 2025) present a framework specifically designed for multi-identity human image animation, introducing "identity-specific embeddings to maintain consistent appearances across individuals and a structural learning mechanism that incorporates depth and surface-normal cues to model human-object interactions." This approach addresses the challenge of maintaining individual identity consistency while enabling realistic interactions between multiple people.

Contact-Based Interaction Modeling: InterAnimate (Lin et al. 2025) focuses specifically on hand-face interactions, which are "essential for emerging biometric authentication systems." The method "simultaneously learns spatio-temporal contact dynamics and biomechanically plausible deformation effects, enabling natural interactions where hand movements induce anatomically accurate facial deformations while maintaining collision-free contact." This detailed modeling of physical contact represents a significant technical advancement in ensuring realistic interaction dynamics.

Zero-Shot Interaction Generation: InterControl (Wang et al. 2024) proposes a framework for zero-shot human interaction generation that adapts "human-wise interactions as pairs of human joints that can be either in contact or separated by a desired distance." This approach enables modeling of interactions involving "an arbitrary number of individuals" without requiring training on specific multi-human configurations, addressing the scalability challenges inherent in multi-person scenarios.

Large-Scale Crowd Dynamics: For crowd-level scenarios, SPDiff (Chen et al. 2024) introduces "a social physics-informed diffusion model" that incorporates Social Force model principles into crowd simulation. The approach "takes both the interactive and historical information of crowds in the current timeframe to reverse the diffusion process," generating realistic pedestrian movement patterns that exhibit both individual behaviors and emergent group dynamics.

Generative Crowd Population: Gen-C (Panayiotou et al. 2025) represents a novel approach that leverages large language models to "generate a limited set of crowd scenarios, which are subsequently expanded and generalized through simulations to construct time-expanded graphs." This hybrid approach demonstrates how different AI technologies can be combined to address the data scarcity challenges in multi-person scenario generation.

These multi-person frameworks collectively demonstrate the progression from basic multi-person generation to sophisticated systems that model social dynamics, physical interactions, and crowd behaviors while maintaining individual identity consistency and realistic coordination patterns. Table 7 summarises the scope, coordination mechanism, and social-dynamics support of the leading multi-person frameworks discussed above.

Quantitative performance across families: As depicted in Table 8, the final column of the table reports the headline metrics published for each method, together with the benchmark on which they were obtained. Several patterns substantiate the qualitative claims made above. Among identity-preserving image-animation methods evaluated on the TikTok benchmark, MagicAnimate (Xu et al. 2024) attains an FVD of 179.1 and SSIM of 0.714, the DisPose (Li et al. 2025) plug-in lowers MimicMotion's FVD from 598.4 to 466.9, and the contact-aware InterAnimate (Lin et al. 2025) reaches the strongest video fidelity (FVD = 122.9, SSIM = 0.785). For pose-guided fashion synthesis, DreamPose (Karras et al. 2023) reports SSIM = 0.885 and FVD = 238.8 on UBC-Fashion. Among physics-informed motion generators, PhysDiff (Yuan et al. 2023) reduces the physics error from 28.37 to 1.46 and ground penetration from 13.08 to 0.87 relative to the kinematics-only MDM baseline on HumanAct12, while FlexMotion (Tashakori et al. 2025) and InterControl (Wang et al. 2024) achieve low FID (0.298 and 0.159) on HumanML3D with substantially reduced foot-skating and joint error. Crucially, because these methods are validated on heterogeneous tasks, datasets, and protocols (e.g., TikTok, UBC-Fashion, HumanML3D, BEHAVE, and ViCo), the reported figures are indicative of each method's own claimed strength rather than directly comparable

Table 7 Multi-Person Interaction Modeling Methods

Method	Interaction Modeling	Physics-Based	Scale	Technical Approach
Dyadic and Small Group Methods				
DIM (Tran et al. 2024)	Conversational dynamics	×	2 persons	Masking and contrastive learning
InterAnimate (Lin et al. 2025)	Hand-face contact	✓	2 persons	Spatio-temporal contact dynamics
Multi-Identity (Wang et al. 2025)	Identity-aware interactions	✓	Small group	Identity embeddings + structural cues
InterControl (Wang et al. 2024)	Joint-based coordination	✓	Variable	Zero-shot joint pair modeling
Two-in-One (Li et al. 2025)	Interactive motion spaces	×	2 persons	Unified latent space design
Crowd and Large-Scale Methods				
SPDiff (Chen et al. 2024)	Social force dynamics	✓	Large crowd	Physics-informed crowd simulation
Gen-C (Panayiotou et al. 2025)	Scenario-based generation	✓	Large crowd	LLM scenario generation + simulation
Specialized Interaction Types				
DBN Model (Mihoub et al. 2016)	Coverbal behavior	×	2 persons	Dynamic Bayesian Networks
Human4DiT (Shao et al. 2024)	360-degree multi-view	×	1 person	Multi-view consistency (single person)

across rows. This very heterogeneity—the absence of a shared human-centric benchmark and protocol—is itself a key finding, and it motivates the unified evaluation paradigm discussed in Sect. 6 and the metric-selection guidance of Sect. 6.4.

5 Literature survey

Video generation using diffusion models has emerged as a rapidly evolving field, with researchers exploring various modalities and control mechanisms to achieve high-quality, temporally consistent video synthesis. This section provides a comprehensive overview of the current state-of-the-art methods, categorized by their input modalities and application domains.

The largest category of approaches combines textual descriptions with additional control inputs to enable more precise video generation. Lee et al. (2023) introduced shape-aware text-driven layered video editing that handles object shape changes through deformation field propagation and attention-based refinement. Cai et al. (2024) presented Generative Rendering, which combines controllable 4D-guided video generation with 2D diffusion models, enabling control over scene geometry, appearance, motion, and camera paths.

Several methods focus on one-shot tuning approaches. Wu et al. (2023) proposed Tune-A-Video, which adapts text-to-image diffusion models for video generation using spatio-temporal attention mechanisms and efficient one-shot tuning strategies. Ceylan et al. (2023) developed Pix2Video, which uses structure-guided image diffusion models for text-guided video editing through progressive frame propagation.

Motion control represents another significant research direction. Zhao et al. (2023) introduced MotionDirector for motion customization in text-to-video generation using dual-path LoRA architectures to decouple appearance and motion learning. Qiu et al. (2024) presented FreeTraj, a tuning-free framework for trajectory control in video diffusion models through noise construction and attention computation guidance.

Recent advances include sophisticated editing frameworks. Geyer et al. (2023) proposed TokenFlow for consistent video editing by enforcing consistency in the diffusion feature space through inter-frame correspondence propagation. Jeong and Ye (2024) developed Ground-A-Video for zero-shot grounded video editing using Cross-Frame Gated Attention and optical flow guidance.

Pure text-to-video generation represents the most straightforward approach, relying solely on textual descriptions. Singer et al. (2022) pioneered Make-A-Video, which leverages paired text-image data and unsupervised video footage without requiring paired text-video data. Ho et al. (2022) introduced Imagen Video, implementing a cascade of video diffusion models with spatial and temporal super-resolution components.

More recent developments include architectural innovations. Yang et al. (2025) presented CogVideoX with expert transformer architectures and 3D Variational Autoencoders for enhanced text-video alignment. Chen et al.

https://doi.org/10.1007/s10462-026-11699-z

Table 8 Comprehensive overview of human-centric video generation methods

Method	Primary Challenge	Input Modalities	Key Innovation	Technical Contribution	Reported Quantitative Results (Benchmark)
Physics-Informed Methods					
PhysDiff (Yuan et al. 2023)	Physical Plausibility	Pose + Physics	Physics projection module	Real-time physics constraint enforcement	HumanAct12: FID 13.27, Acc. 0.956; phys-err 1.46 vs. 28.37 and penetration 0.87 vs. 13.08 (MDM)
BioVAE (Kang et al. 2025)	Biomech. accuracy	Motion (SMPL)	Biomechanical residual VAE	Biomechanics-guided motion refinement & estimation	3DPW: PA-MPJPE 36.1, accel. 5.55 (vs. 6.29); HumanML3D FID 0.071
FlexMotion (Tashakori et al. 2025)	Physics + efficiency	Joint + contact forces	Latent physics embedding	Inference-time physics-free generation	HumanML3D: R-Prec 0.757, FID 0.298; foot-skate 0.61, penetration 4.95 vs. 11.29 (MDM)
InterDiff (Xu et al. 2023)	Human-object physics	3D pose + objects	Two-stage correction	Physics-informed interaction modeling	BEHAVE: HOI prediction error 84/120 vs. 117/158 without physics correction
Identity-Preserving Methods					
MagicAnimate (Xu et al. 2024)	Identity consistency	Image + Pose	Parallel processing paths	Temporal + appearance encoding	TikTok: SSIM 0.714, PSNR 18.22, LPIPS 0.239; FVD 179.1, FID-VID 21.8 (vs. 292.8 DisCo)
Magic Mirror (Zhang et al. 2025)	Identity vs. motion	Image + Pose	Dual-branch feature extractor	Identity-structure disentanglement	ID-Sim 0.922, dynamic degree 0.705, preference 7.32/10; 74.6% user-study win-rate
EchoVideo (Wei et al. 2025)	Copy-paste artifacts	Image + Text	Multi-modal identity fusion	Semantic identity representation	ID-Avg 0.516 (vs. 0.414 ConsisID), dynamic degree 0.955, inception dist. 176.5
MotionCharacter (Fang et al. 2024)	Flexible identity control	Image + attributes	Region-aware losses	Selective identity preservation	Face-Sim 0.609, DOVER 0.869, motion smooth. 0.998, CLIP-T 0.227
DisPose (Li et al. 2025)	Body shape adaptation	Image + sparse pose	Motion field generation	Dense-sparse pose disentanglement	TikTok (+MimicMotion): FVD 466.9 (vs. 598.4), SSIM 0.781, subject-cons. 95.7
Pose-Guided Control Methods					
DreaMoving (Feng et al. 2023)	Motion control	Text + Image + Pose	Video ControlNet	Multi-modal conditioning framework	Qualitative evaluation only; no benchmark metrics reported
Follow Your Pose (Ma et al. 2024)	Foundation model adaptation	Text + Pose	Two-stage training	Temporal self-attention mechanisms	Highest CLIP frame-consistency and pose accuracy vs. ControlNet/Tune-A-Video
MimicMotion (Zhang et al. 2024)	Pose uncertainty	Pose + Confidence	Confidence-aware guidance	Regional loss amplification	TikTok: FVD 594, SSIM 0.795, PSNR 20.1, FID-VID 9.3
DreamPose (Karras et al. 2023)	Fashion animation	Image + Pose	Domain-specific adaptation	Fashion-optimized architectures	UBC-Fashion: SSIM 0.885, LPIPS 0.068, FID 13.04, FVD 238.8
Multi-Person Frameworks					
InterControl (Wang et al. 2024)	Scalable interactions	Joint pairs	Zero-shot generation	Arbitrary person number handling	HumanML3D: FID 0.159, R-Prec 0.671; avg. joint err. 0.05 m, foot-skate 0.073
Multi-Identity (Wang et al. 2025)	Multi-person identity	Multiple images	Identity embeddings	Structural learning mechanisms	SSIM 0.691, PSNR 20.69, LPIPS 0.233; FVD 878, FID 30.6
SPDiff (Chen et al. 2024)	Crowd dynamics	Trajectory + History	Social physics modeling	Interactive crowd simulation	GC/UCY crowds: 6.5-−13.5% lower MAE/OT/MMD than prior baselines
InterAnimate (Lin et al. 2025)	Contact modeling	Hand-face regions	Contact dynamics learning	Biomechanical deformation effects	FID 23.2, SSIM 0.785, PSNR 23.4; FVD 122.9 (vs. 176.2 StableAnimator)
DIM (Tran et al. 2024)	Conversational behavior	Audio + 3D face	Dyadic context modeling	Contrastive learning for coordination	ViCo listener: feature dist. 24.2, MSE 0.62; top user-study preference

The final column reports the headline quantitative results published for each method, together with the benchmark on which they were obtained

Because the methods target different tasks, datasets, and evaluation protocols, these figures are indicative of each method's own reported strength rather than directly comparable across rows

(2024) proposed GenTron, adapting Diffusion Transformers from class to text conditioning while scaling model parameters for improved visual quality.

Qing et al. (2024) introduced hierarchical spatio-temporal decoupling through HiGen, separating spatial reasoning and temporal reasoning using unified denoisers. Lee et al. (2024) developed Grid Diffusion Models that represent videos as grid images to enable generation with fixed memory requirements regardless of frame count.

Combining textual and visual inputs enables more controlled video generation. Zhang et al. (2023) developed I2VGen-XL using cascaded diffusion models with hierarchical encoders for semantic coherence and detail enhancement. Chen et al. (2023) introduced VideoCrafter1 as an open-source framework supporting both text-to-video and image-to-video generation with cinematic quality.

Jiang et al. (2024) proposed VideoBooth for diffusion-based video generation with image prompts, employing coarse-to-fine visual embeddings and attention injection modules. Wang et al. (2024) presented MicroCinema using a divide-and-conquer strategy that separates text-to-image and image-text-to-video generation stages.

Recent innovations include autoregressive approaches. Gao et al. (2024) introduced ViD-GPT, bringing causal generation to video diffusion models with past frames as prompts for future frame generation. Hacohen et al. (2024) developed LTX-Video for real-time video generation using holistic transformer-based latent diffusion with high compression ratios.

Advanced control mechanisms combine text-image inputs with additional modalities. Molad et al. (2023) presented Dreamix for general video editing using mixed objectives that jointly finetune with temporal attention and masking. Wang et al. (2023) developed VideoComposer for compositional video synthesis with motion controllability using motion vectors and spatio-temporal condition encoders.

Zhang et al. (2025) introduced Tora, the first trajectory-oriented diffusion transformer framework that integrates textual, visual, and trajectory conditions using hierarchical spacetime motion patches. Chen et al. (2023) proposed SEINE for short-to-long video generation with generative transition and prediction capabilities.

Bar-Tal et al. (2024) presented Lumiere, implementing a Space-Time U-Net architecture that generates entire temporal durations through single-pass processing. Wang et al. (2024) developed Boximator for fine-grained motion control using hard and soft box constraints.

Image-centric approaches focus on leveraging visual information for video generation. Blattmann et al. (2023) introduced Video LDMs by applying latent diffusion paradigms to high-resolution video generation through temporal alignment of pretrained image models. Ni et al. (2023) proposed conditional image-to-video generation using Latent Flow Diffusion Models that synthesize optical flow sequences in latent space.

Xu et al. (2024) developed CamCo for camera-controllable 3D-consistent image-to-video generation using Pl cker coordinates and epipolar attention modules. Liu et al. (2024) presented MarDini, combining masked autoregression advantages with diffusion models through asymmetric network design.

Human-centric video generation represents a specialized but crucial application domain. For text-driven human motion, Chen et al. (2023) introduced Motion Latent-based Diffusion (MLD) that performs diffusion processes in motion latent space rather than raw sequences. Zhou and Wang (2023) proposed UDE (Unified Driving Engine) for generating human motion from natural language or audio sequences using motion quantization and transformer-based diffusion.

Image-pose human motion synthesis has seen significant advances. Hu (2024) presented Animate Anyone for consistent and controllable character animation using ReferenceNet for appearance preservation and pose guiders for motion control. Xu et al. (2024) developed MagicAnimate for temporally consistent human image animation using appearance encoders and video fusion techniques.

Wang et al. (2024a) introduced DisCo for realistic human dance generation with disentangled control, improving generalizability across poses and human subjects. Wang et al. (2024) proposed LEO (Generative Latent Image Animator) using flow-based image animation and Latent Motion Diffusion Models for spatio-temporal coherency.

Advanced pose-based approaches include (Zhang et al. 2024), which presented MimicMotion using confidence-aware pose guidance and progressive latent fusion for long video generation. Li et al. (2025) developed DisPose for disentangling pose guidance into motion field guidance and keypoint correspondence.

Physics-informed video generation addresses the challenge of generating physically plausible motions. Yuan et al. (2023) introduced PhysDiff, incorporating physical constraints through motion imitation in physics simulators. Kulkarni et al. (2024) proposed NIFTY (Neural Interaction Fields for Trajectory sYnthesis) for guided human motion synthesis with object interactions.

Xu et al. (2023) presented InterDiff for generating 3D human-object interactions with physics-informed diffusion, addressing dynamic objects and whole-body motion modeling. Huang et al. (2023) developed SceneDiffuser for scene-aware, physics-based, and goal-oriented generation across multiple 3D scene understanding tasks. Several specialized applications have emerged. Ruan et al. (2023) introduced MM-Diffusion for joint audio-video generation using coupled denoising autoencoders and random-shift attention blocks. Karras et al. (2023) presented DreamPose for fashion image-to-video synthesis using architectural changes and temporal consistency techniques.

Tian et al. (2024) developed EMO for expressive portrait videos using direct audio-to-video synthesis without intermediate 3D models. He et al. (2024) proposed co-speech gesture video generation via motion-decoupled diffusion models using TPS transformations and temporal correlation learning.

Recent work has focused on computational efficiency. Zhao et al. (2024) introduced ViDiT-Q for efficient quantization of diffusion transformers, achieving significant memory optimization and latency speedup. Tian et al. (2024) presented QVD for post-training quantization specifically designed for video diffusion models with scattered channel range integration.

Zhang et al. (2024) developed SF-V for single forward video generation using adversarial training to fine-tune pretrained models, achieving substantial speedup compared to existing methods. Oshima et al. (2024) explored state-space models as alternatives to attention layers for longer video generation with linear memory consumption.

The field continues to evolve with emerging trends including autoregressive video generation (Sun et al. 2025), zero-shot 4D video generation (Park et al. 2025), and improved controllability mechanisms (Peng et al. 2025). Key challenges remain in achieving long-term temporal consistency, handling complex multi-object interactions, and balancing computational efficiency with generation quality.

The comprehensive taxonomy presented in Table 9 demonstrates the diverse approaches researchers have taken to address these challenges, from pure text-based generation to sophisticated multimodal systems incorporating pose sequences, physical constraints, and specialized human motion synthesis capabilities. As the field progresses, we anticipate continued innovation in control mechanisms, architectural designs, and application-specific optimizations.

5.1 Comparative discussion: method-to-application guidance

The breadth of approaches reviewed above makes it difficult to judge, for a given application, which method to pick. We summarise the practical trade-offs per method family below, translating the architectural descriptions into operational recommendations.

Pure text-to-video methods: (Make-A-Video, Imagen Video, CogVideoX, GenTron) are the right default when the user only has a natural-language prompt and the output length is short (typically ≤ 4 s at present). Their strengths are mature cascaded super-resolution pipelines and strong semantic grounding; their weaknesses are limited fine control over identity, motion trajectory, and long-range consistency, and high inference cost.

Image- or video-conditioned methods: (Stable Video Diffusion, I2VGen-XL, VideoCrafter1, MarDini) dominate when a reference frame is available. They yield stronger appearance fidelity than pure T2V at the cost of reduced motion diversity, and they are the preferred substrate for downstream human-centric pipelines because they accept a reference identity directly.

Table 9 Video Generation and/or Editing Methods with Coded Categories: T (Text Input), I (Image Input), P (Pose Sequence Input), O (Other Inputs: Video, Audio, Depth, etc.), HM (Human Motion Focus), PC (Physical Constraints), MM (Multi-Modal Integration), Q (Quantization as 3.2.2/Optimization)

Category Code	Article	T	I	P	O	HM	PC	MM	Q
T-O	(Cai et al. 2024; Wu et al. 2023; Zhao et al. 2023; Wang et al. 2023; Geyer et al. 2023; Li et al. 2025; Yoon et al. 2023; Ma et al. 2025; Yang et al. 2024; Ren et al. 2024; Qiu et al. 2024; Yang et al. 2025; Jain et al. 2024; Wang et al. 2024; Xing et al. 2024; Jeong et al. 2024; Dhesikan and Rajmohan 2023; Li et al. 2025; Lv et al. 2024)	✓	×	×	✓	×	×	✓	×
T	(Luo et al. 2023; Ho et al. 2022; Liu et al. 2023; Wang et al. 2023; Hong et al. 2024; Wang et al. 2023; Zhang et al. 2025; Bao et al. 2024; Cho et al. 2024; Zhang et al. 2025; Chen et al. 2024, 2024; Fei et al. 2024; Qing et al. 2024; Ge et al. 2023; Zhang et al. 2024; Li et al. 2024; Skorokhodov et al. 2024; Yuan et al. 2024)	✓	×	×	×	×	×	×	×
T-Q	(Zhao et al. 2024)	✓	×	×	×	×	×	×	✓
TI	(Molad et al. 2023; Zhou et al. 2023; Singer et al. 2022; Qin et al. 2024; Wang et al. 2024; Li et al. 2023; Zhang et al. 2023; Chen et al. 2023; Guo et al. 2024; Jiang et al. 2024; Wang et al. 2024; Gao et al. 2024; Hacohen et al. 2024; Liu et al. 2023; Liang et al. 2024; Song et al. 2023; Oh et al. 2024; Gupta et al. 2023; Chen et al. 2023; Lee et al. 2024; Liu et al. 2023; Zhang et al. 2023; Tian et al. 2024; Qi et al. 2023)	✓	✓	×	×	×	×	✓	×
TI-O	(Wu et al. 2023a; Wang et al. 2023; Wu et al. 2023b; Guo et al. 2024; Zhang et al. 2025; Bar-Tal et al. 2024; Feng et al. 2024; Zeng et al. 2024; Wang et al. 2024; Hu and Xu 2023; Liew et al. 2023; Jeong and Ye 2024; Cong et al. 2024; Li et al. 2024; Chu et al. 2024; Ouyang et al. 2024; Blattmann et al. 2023; Ceylan et al. 2023; Chai et al. 2023; Esser et al. 2023; Lee et al. 2023; Yu et al. 2024; Kuang et al. 2024; Wei et al. 2024; Zheng et al. 2024)	✓	✓	×	✓	×	×	✓	×
TI-HM	(He et al. 2024; Ma et al. 2024)	✓	✓	×	×	✓	×	✓	×
TI-O-HM	(Fang et al. 2024)	✓	✓	×	✓	✓	×	✓	×
TP-HM	(Kim et al. 2023)	✓	×	✓	×	✓	×	✓	×
TP-O-HM	(Yang et al. 2024; Kim et al. 2023; Qin et al. 2023; Zhou and Wang 2023)	✓	×	✓	✓	✓	×	✓	×
TP-HM-PC	(Yuan et al. 2023; Gao et al. 2024)	✓	×	✓	×	✓	✓	✓	×
TIP-HM	(Feng et al. 2023; Ma et al. 2024; Kim et al. 2023; Xu et al. 2024)	✓	✓	✓	×	✓	×	✓	×
TIP-O	(Peng et al. 2025; Khachatryan et al. 2023)	✓	✓	✓	✓	×	×	✓	×
TI-O-Q	(Tian et al. 2024)	✓	✓	×	✓	×	×	✓	✓
I	(Yang et al. 2023; Harvey et al. 2022; Ni et al. 2023)	×	✓	×	×	×	×	×	×
I-O	(Xu et al. 2024; Rigter et al. 2024; Liu et al. 2024; Park et al. 2025; Lee et al. 2025)	×	✓	×	✓	×	×	✓	×
IP-HM	(Hu 2024; Karras et al. 2023; Wang et al. 2024a, b)	×	✓	✓	×	✓	×	✓	×
IP-O	(Niu et al. 2024)	×	✓	✓	✓	×	×	✓	×
IP-O-HM	(Men et al. 2024; Shao et al. 2024; Zhang et al. 2024; Li et al. 2025; Wang et al. 2025; Lin et al. 2025)	×	✓	✓	✓	✓	×	✓	×
P-HM	(Du et al. 2023)	×	×	✓	×	✓	×	×	×
P-O-HM-PC	(Kulkarni et al. 2024; Xu et al. 2023; Huang et al. 2023; Tseng et al. 2023)	×	×	✓	✓	✓	✓	✓	×
T-O-HM	(Chen et al. 2023)	✓	×	×	✓	✓	×	✓	×
TP-O-HM-PC	(Dabral et al. 2023)	✓	×	✓	✓	✓	✓	✓	×
I-HM	(Wang et al. 2024)	×	✓	×	×	✓	×	×	×
I-O-HM	(Tian et al. 2024; Das and Das 2024; He et al. 2024; Stypułkowski et al. 2024)	×	✓	×	✓	✓	×	✓	×
O	(Oshima et al. 2024; Sun et al. 2025; Voleti et al. 2022; Yariv et al. 2024)	×	×	×	✓	×	×	×	×
O-MM	(Ruan et al. 2023)	×	×	×	✓	×	×	✓	×

https://doi.org/10.1007/s10462-026-11699-z

Pose- or trajectory-guided methods: (UniAnimate, MotionCtrl, HumanDiT, VividPose) should be selected whenever motion must be explicitly controllable—avatar animation, fashion, dance generation. The trade-off is sensitivity to body-shape mismatch between reference and target, which DisPose (Li et al. 2025) explicitly addresses through sparse-to-dense motion fields.

Physics-informed methods: (PhysDiff, BioVAE, FlexMotion, InterDiff; see Table 5) are necessary when biomechanical plausibility is evaluated (rehabilitation, sports, robotics). They trade increased inference cost (simulation in the loop) for principled physical guarantees. FlexMotion, which embeds physics in the latent space, offers a middle ground and is currently the best choice for real-time applications.

Identity-preserving architectures: (MagicAnimate, Magic Mirror, EchoVideo, MotionCharacter; see Table 6) should be favoured when a specific person must be animated. Transformer-based adapter approaches (Magic Mirror) outperform earlier ReferenceNet-style encoders for long sequences but require more training data.

Efficiency-oriented methods: (ViDiT-Q, QVD, MobileVD, SF-V) are complementary: they can be composed with any of the above families post-training. For human-centric deployment, combining a pose-guided backbone with MobileVD-style temporal block pruning and QVD-style post-training quantization is currently the only route to sub-second inference at acceptable quality.

The overall pattern is that *no single method dominates*: method selection should be driven by (i) availability of the conditioning signal (text / image / pose / audio), (ii) the required control granularity, (iii) the latency budget, and (iv) whether biomechanical plausibility is auditable. This contingent guidance is reflected in our metric-to-application mapping in Sect. 6.

6 Evaluation methodology

Our evaluation methodology focuses on assessing three key aspects of our framework: multi-modal integration effectiveness, human motion realism, and computational efficiency after quantization. The assessment framework is designed to rigorously validate our model's ability to handle multiple input modalities while maintaining realistic human motion and computational efficiency.

6.1 Datasets

The selection of appropriate datasets is crucial for evaluating our multi-modal video generation framework. Given the complexity of our approach, which combines text, image, and pose sequence inputs, we require datasets that provide rich annotations across multiple modalities while maintaining high-quality human motion data. Table 10 provides a comprehensive overview of our selected datasets and their key characteristics.

6.1.1 Primary evaluation datasets

Our primary evaluation relies on three key datasets that offer comprehensive coverage of our model's requirements. HumanML3D (Guo et al. 2022) serves as our foundational dataset, providing 14,616 motions paired with 44,970 text descriptions, enabling robust evaluation of text-to-motion generation capabilities and physiological constraint enforcement. This dataset is particularly valuable for assessing the model's ability to maintain natural human motion patterns.

Fashion-Text2Video (Jiang et al. 2023) complements our evaluation framework with 600 fashion show videos accompanied by detailed motion labels. This dataset is especially useful for evaluating human motion in controlled settings, offering a consistent environment for assessing temporal coherence and multi-modal integration. The structured nature of fashion shows provides an ideal testbed for evaluating pose sequence accuracy and appearance consistency.

Table 10 Overview of selected evaluation datasets

Dataset	Type	Size	Key Features	Primary Evaluation Purpose
HumanML3D	Motion + Text	14,616 motions 44,970 descriptions	Text-motion pairs, diverse actions	Motion naturalness, text-motion alignment
Fashion-Text2Video	Video + Text	600 videos	Fashion show sequences, motion labels	Temporal coherence, controlled motion evaluation
iPER	Video	206 videos	Multiple outfits, pose variations	Appearance consistency, pose-guided generation
VATEX	Text-Video	241,000 pairs	Rich action descriptions, diverse scenarios	Multi-modal coherence, text understanding
A2D	Video + Action	Various	Action annotations, actor labeling	Action-text alignment, motion synthesis
AIST++	Motion	1,408 sequences	Dance sequences, 10 genres	Complex motion evaluation, temporal consistency
Human3.6M	Motion Capture	3.6M frames	Professional mocap, 15 actions	Physiological constraints, motion accuracy
DeepFashion Video	Video + Images	Various	Clothing attributes, high quality annotations, 15 actions	Appearance preservation, identity consistency, motion accuracy
MPII Human Pose	Image + Pose	40,000 images	Detailed pose annotations, diverse activities, 10 genres	Pose-guided appearance consistency, temporal consistency

The iPER dataset (Liu et al. 2019) rounds out our primary evaluation set with 206 videos of human motions in different clothing conditions. This dataset is particularly valuable for testing appearance consistency across varying poses and outfits, providing a robust evaluation of our model's ability to maintain visual coherence while adapting to different human poses.

6.1.2 Multi-modal validation datasets

For validating our multi-modal integration capabilities, we employ two complementary datasets. VATEX (Wang et al. 2019), with its extensive collection of 241,000 text-video pairs, provides rich textual descriptions of human actions, enabling comprehensive evaluation of text-motion alignment and multi-modal coherence. The Actor-Action (A2D) dataset (Gavrilyuk et al. 2018) supplements this with its focus on specific actions and actors, offering ground truth for action-text alignment evaluation.

6.1.3 Reference image extraction

To strengthen our evaluation of appearance consistency and identity preservation, we incorporate two specialized reference image datasets. DeepFashion Video (Liu et al. 2016) provides extensive video sequences paired with high-quality annotated images, featuring detailed clothing attributes and pose annotations. This dataset is instrumental in evaluating the model's ability to maintain consistent appearance and identity across generated video frames while preserving clothing details. MPII Human Pose Dataset (Andriluka et al. 2014) contributes over 40,000 images with precise pose annotations across diverse human activities. This dataset serves as a robust benchmark for assessing pose-guided appearance consistency and evaluating the model's capability to maintain visual fidelity across varying poses.

6.1.4 Specialized test sets

Our specialized evaluation employs AIST++ (Li et al. 2021) and Human3.6M (Ionescu et al. 2014) datasets for detailed motion analysis. AIST++ provides 1,408 dance motion sequences across 10 genres, offering complex human movements ideal for testing our constraint system. Human3.6M, with its 3.6 million frames of

professional motion capture data, serves as our gold standard for evaluating physiological constraint adherence and motion naturalness.

As shown in Table 10, each dataset has been carefully selected to evaluate specific aspects of our framework while ensuring comprehensive coverage of our model's capabilities. This diverse yet focused selection enables thorough evaluation of multi-modal integration, human motion synthesis, and the effects of quantization on generation quality.

6.2 Evaluation metrics

Our evaluation framework employs a comprehensive set of metrics designed to assess the three critical aspects of our multi-modal video generation system: visual quality, motion realism, and computational efficiency. All metrics will be computed across multiple generated sequences and averaged to ensure robust evaluation.

6.2.1 Video quality assessment

For visual quality assessment, we employ a complementary set of metrics that evaluate both frame-wise quality and temporal coherence:

1. Fr chet Video Distance (FVD) (Unterthiner et al. 2019):

$$FVD = \|\mu_r - \mu_g\|^2 + Tr(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}) \tag{33}$$

where μ_r, Σ_r represent the mean and covariance of real video features, and μ_g, Σ_g represent those of generated videos. Lower FVD scores indicate better video quality and temporal consistency.

2. Peak Signal-to-Noise Ratio (PSNR):

$$PSNR = 20 \log_{10} \left(\frac{MAX_I}{\sqrt{MSE}} \right) \tag{34}$$

where MAX_I is the maximum possible pixel value. Higher PSNR values indicate better frame-level quality.

3. Structural Similarity Index (SSIM):

$$SSIM(x, y) = \frac{(2\mu_x \mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{35}$$

where C_1 and C_2 are constants to stabilize division. SSIM ranges from -1 to 1, with 1 indicating perfect structural similarity.

6.2.2 Motion realism metrics

1. Average Euclidean Distance (AED): AED measures the spatial discrepancy between the generated pose/keypoints and the ground truth pose/keypoints across all frames.

$$\text{AED} = \frac{1}{N} \sum_{i=1}^{N} \sqrt{\sum_{j=1}^{K} \left\| \mathrm{p}_j^i - \hat{\mathrm{p}}_j^i \right\|^2} \tag{36}$$

Where:

- N: Total number of frames in the sequence.
- K: Number of keypoints (e.g., body joints).
- p_j^i: Ground truth position of the j-th keypoint in frame i.

- $\hat{p}_j^i$: Generated position of the j-th keypoint in frame i.

2. End-Point Error (EPE): EPE evaluates the motion alignment by calculating the displacement error between the predicted and ground truth key-point trajectories.

$$\text{EPE} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\sum_{j=1}^{K}\left\|(p_j^{i+1} - p_j^i) - (\hat{p}_j^{i+1} - \hat{p}_j^i)\right\|^2} \tag{37}$$

Where:

- $(p_j^{i+1} - p_j^i)$: Ground truth motion vector of the j-th keypoint between frames i and $i+1$.
- $(\hat{p}_j^{i+1} - \hat{p}_j^i)$: Generated motion vector of the j-th keypoint.

3. Temporal Consistency (TC): TC measures the smoothness and continuity of the generated motion over time by comparing changes between consecutive frames.

$$\text{TC} = \frac{1}{N-1}\sum_{i=1}^{N-1}\sqrt{\sum_{j=1}^{K}\left\|(\hat{p}_j^{i+1} - \hat{p}_j^i) - (\hat{p}_j^i - \hat{p}_j^{i-1})\right\|^2} \tag{38}$$

Where:

- Temporal derivatives of generated motion are used to calculate consistency between adjacent frames.

6.2.3 Computational efficiency metrics

Our evaluation of computational efficiency encompasses both model compression and runtime performance metrics.

1. The Model Size Reduction (MSR) quantifies the effectiveness of our quantization approach:

$$MSR = \left(1 - \frac{S_{\text{quantized}}}{S_{\text{original}}}\right) \times 100\% \tag{39}$$

2. Generation speed is measured through Inference Time (IT), calculated as frames per second:

$$IT = \frac{N_{\text{frames}}}{T_{\text{generation}}} \tag{40}$$

3. Memory efficiency is assessed using the Memory Utilization Reduction (MUR) metric:

$$MUR = \left(1 - \frac{M_{\text{quantized}}}{M_{\text{original}}}\right) \times 100\% \tag{41}$$

4. Finally, we introduce a Quality-Performance Trade-off (QPT) metric that combines quality and efficiency measures:

$$QPT = \alpha \cdot Q_{\text{norm}} + (1-\alpha) \cdot S_{\text{norm}} \tag{42}$$

where α is an application-dependent weighting factor, Q_{norm} represents normalized quality metrics, and S_{norm} represents normalized speed metrics. This comprehensive set of metrics enables thorough evaluation of our framework's performance across all critical dimensions, from visual quality to computational efficiency.

6.3 Caveats on efficiency comparability

Efficiency gains reported in the literature—including the 523× reduction achieved by MobileVD (Yahia et al. 2024) and large speed-ups reported for QVD (Tian et al. 2024), ViDiT-Q (Zhao et al. 2024), and SF-V (Zhang et al. 2024)—should be read with three standard caveats that we make explicit here in the interest of fair comparison.

(i) Baseline heterogeneity: Reported multipliers depend on the specific baseline (SVD vs. Imagen Video vs. LDM), the resolution (256^2, 512^2, 768^2), and the number of sampled frames. A 10× speed-up on a 1024^2 long video is often harder than a 20× speed-up on a 256^2 short clip.

(ii) Quality metrics are not directly comparable across studies: FVD is sensitive to the feature-extractor checkpoint; FID varies with the reference distribution; human-preference studies differ in participant pool, prompt set, and question design. A method reporting "FVD 149 vs. 171" on one benchmark cannot be meaningfully compared to "FVD 185 vs. 210" on another.

(iii) Hardware profile: Speed-ups measured on H100s do not generalise to consumer GPUs, and compiler-level optimisations (TensorRT, torch.compile) often account for a non-trivial fraction of the reported gain, independent of the method's algorithmic contribution.

For the MIME-Vid evaluation plan, we commit to reporting (a) the baseline checkpoint and resolution, (b) FVD on a fixed I3D extractor checkpoint, (c) inference-time measurements on a single stated GPU tier, and (d) ablations against pure quantization and pure pruning baselines, so that the computational savings can be attributed to specific components rather than to their combination. We encourage future work to adopt this reporting convention.

6.4 Metric-to-application guideline

The evaluation metrics introduced above are not equally appropriate for every application. A common pitfall is to rank methods by FVD alone when the target application (e.g., medical movement demonstration) is in fact dominated by physical-plausibility concerns. Table 11 provides a structured guideline mapping application scenarios to the metrics that practitioners should prioritise. The mapping is not exhaustive, but covers the dominant use-cases identified across the surveyed literature.

The central recommendation, reflecting the cautions in Sect. 6.3, is that reviewers and practitioners should select the primary metric(s) based on *the decision the evaluation must support*, rather than adopting FVD as a

Table 11 Mapping of evaluation metrics to application scenarios. Primary metrics are those that most directly reflect the decision criterion for the application; secondary metrics should be reported but rarely decide method selection alone

Application Scenario	Primary Metrics	Secondary Metrics / Notes
Text-to-video content creation	FVD, CLIP-Score, Human preference	PSNR, SSIM (less informative at high resolutions)
Avatar animation (identity critical)	Face-ID similarity, Temporal Consistency (TC), human preference	FVD, AED on facial landmarks
Fashion / virtual try-on	SSIM, LPIPS, Identity similarity	FVD, appearance consistency across frames
Dance / sports / motion synthesis	AED, EPE, Temporal Consistency (TC), joint-angle validity	FVD, physics-simulator score
Medical / rehabilitation demonstration	Joint-angle validity, contact-force plausibility, biomechanical score	AED, TC; FVD only as sanity check
Multi-person / social interaction	Spatial coordination error, interaction synchronisation, per-identity similarity	FVD, crowd-level behavioural metrics
Efficiency-constrained deployment (mobile, real-time)	IT (frames/s), MSR, MUR, QPT at fixed quality threshold	FVD at the achieved speed; comparison must share hardware profile
Research benchmarking	FVD, PSNR, SSIM, CLIP-Score, Temporal Consistency	All of the above; report on standard splits

universal proxy. This guideline is the structural complement to our metric definitions above and to the efficiency caveats in the preceding subsection.

6.5 Evaluation metrics and human perception: strengths, weaknesses, and biases

The quantitative results compiled in Table 8 are only as meaningful as the metrics that produce them, and for human-centric video the gap between a metric's numerical optimum and human perceptual judgement is often decisive. We therefore summarise, for the metrics that recur throughout the surveyed literature, what each actually measures, where it aligns with human perception, and the systematic biases a reader should keep in mind when interpreting reported numbers.

Pixel-fidelity metrics (PSNR, SSIM, L1): These reward per-pixel or per-window agreement with a reference frame and are stable and cheap to compute, which is why they dominate fashion and image-animation benchmarks (e.g. DreamPose, MagicAnimate in Table 8). Their weakness is that they correlate weakly with perceived quality once resolution and motion increase: a temporally smooth but blurry result can outscore a sharp result with slight misalignment, and neither captures motion naturalness. They are best read as necessary-but-not-sufficient sanity checks rather than as arbiters of perceptual quality.

Distribution metrics (FID, FVD, FID-VID): FVD is the de-facto standard for video and correlates better with holistic human judgement than pixel metrics because it compares distributions of spatio-temporal features. However, it carries three well-documented biases: it is sensitive to the I3D feature-extractor checkpoint, to clip length and frame rate, and to the reference set, so absolute values are not comparable across papers (the reason the final column of Table 8 is annotated with its benchmark and labelled as not cross-comparable). FVD also tends to reward overall realism while remaining relatively insensitive to localised but perceptually salient artifacts such as a single distorted hand or face.

Perceptual and semantic metrics (LPIPS, CLIP-Score): LPIPS, computed in a learned feature space, aligns more closely with human similarity judgements than PSNR/SSIM, but its scores are backbone-dependent and shift with the network used. CLIP-Score measures text–frame semantic alignment, not temporal coherence or anatomical plausibility, so a method can score well while producing flickering or physically implausible motion; it should never be used as a standalone quality measure for human-centric output.

Identity and motion-specific metrics: Face-identity similarity (e.g. ArcFace cosine) is central to the identity-preserving methods in Table 8, yet it is vulnerable to a characteristic bias: a near-static "copy-paste" of the reference face maximises the score while destroying expressiveness and motion—precisely the failure EchoVideo was designed to avoid. This is a clear case where optimising the metric and satisfying the human viewer diverge. Likewise, physics- and motion-oriented scores (foot-skating ratio, penetration, joint-angle validity, AED) capture plausibility dimensions that distribution metrics miss, but they say nothing about appearance, so methods such as PhysDiff that excel on physics error can still look perceptually stiff.

Implication: No single automatic metric is an unbiased proxy for human perception of human-centric video, and the dominant metrics are biased in *different and partly opposite* directions—pixel metrics over-reward smoothness, distribution metrics under-weight local artifacts, semantic metrics ignore dynamics, and identity metrics can reward stasis. The practical consequence, consistent with Table 11, is that human-centric evaluation should report a small *panel* of complementary metrics chosen for the target decision, and should treat a well-designed human-preference study—with a disclosed participant pool, prompt set, and question design—as the final arbiter rather than any single scalar.

6.6 Comparative analysis

Our work builds upon and compares against several pioneering approaches in video generation from different modalities:

6.6.1 Text-to-video approaches

- Make-A-Video (Singer et al. 2022): Leverages text-to-image model knowledge for video generation without requiring text-video paired data. It uses spatial-temporal decomposition in the generation process.
- Imagen Video (Ho et al. 2022): A cascade of video diffusion models that enables high-definition video generation through a sequence of temporal and spatial super-resolution models.

6.6.2 Image-conditioned approaches

- VideoComposer (Wang et al. 2023): Introduces a compositional generation approach that decomposes videos into spatial content and temporal dynamics components.
- Stable Video Diffusion (Blattmann et al. 2023): Introduces a latent video diffusion model that scales to large datasets, enabling high-resolution image-to-video generation.

6.6.3 Motion-guided approaches

- Dreampose (Karras et al. 2023): A diffusion-based method that generates animated fashion videos from still images.
- Animate anyone (Hu 2024): A diffusion-based framework that transforms still character images into consistent and controllable animated videos.

6.6.4 DreaMoving: a human video generation framework based on diffusion models

- DreaMoving (Feng et al. 2023): A diffusion-based controllable video generation framework capable of producing high-quality customized human videos.

7 Research gaps and future directions

7.1 Research gaps in human-centric motion generation

Generating realistic, controllable, and expressive human motion represents a cornerstone challenge in computer graphics and AI, with significant implications for animation, virtual reality, robotics, and human-computer interaction (Janson et al. 2024; Tashakori et al. 2025; Yan et al. 2024). Despite recent advances in diffusion models for this task (Tevet et al. 2022; Yuan et al. 2023), several critical research gaps persist that limit the realism and applicability of generated human motion.

7.1.1 Physiological constraints and motion naturalness

A primary limitation of current human motion generation models is the frequent production of movements that violate fundamental laws of physics or human biomechanics (Tashakori et al. 2025; Yuan et al. 2023; Kang et al. 2025). Common artifacts include:

- Characters appearing to float above the ground
- Feet sliding unnaturally during locomotion

- Limbs penetrating objects or the ground
- Impossible joint rotations
- Lack of convincing weight and dynamics

Standard video or motion diffusion models typically lack innate knowledge of physics or the complex constraints governing human movement (Yuan et al. 2023; Kang et al. 2025). Even when learning from motion capture data, this data may not perfectly capture all physical nuances or cover the full spectrum of humanly possible motions (Janson et al. 2024). Recent work has begun addressing this gap through physics-guided or biomechanically-informed diffusion models. PhysDiff (Yuan et al. 2023) introduces a physics-based motion projection module using simulation to enforce physical constraints during the diffusion process. BioVAE (Kang et al. 2025) introduces a biomechanics-guided residual approach, using a biomechanical variational autoencoder to refine generated motion and improve physical plausibility across generation and estimation. FlexMotion (Tashakori et al. 2025) incorporates joint locations, contact forces, and muscle activations within a latent diffusion framework.

7.1.2 Identity consistency in animation

Another significant challenge is maintaining the visual identity of a specific person, provided via a reference image, throughout a generated animation sequence (Show Lab 2024; Xu et al. 2024; Zhang et al. 2025). This requires effectively disentangling static appearance features from dynamic motion information and ensuring consistency across all generated frames. Challenges in this area include:

- Architectural differences between models optimized for static image generation and those designed for video generation (Zhang et al. 2025)
- Limited availability of high-quality image-video pairs that strictly maintain identity consistency (Zhang et al. 2025)
- Identity degradation during complex or dynamic movements

Recent frameworks targeting this problem include MagicAnimate (Xu et al. 2024), which employs a dedicated video diffusion model for temporal encoding and a novel appearance encoder, and Magic Mirror (Zhang et al. 2025), which utilizes a dual-branch facial feature extractor and a lightweight adapter within a Video Diffusion Transformer architecture.

7.1.3 Nuanced motion style control

While current research focuses heavily on generating specific actions (e.g., "walking," "sitting") and preserving identity, the ability to control the *style* or *manner* in which an action is performed remains relatively underexplored. Generating a "sad walk" versus an "energetic walk," or controlling specific dance styles based on textual descriptions, represents a more nuanced level of control that is currently difficult to achieve reliably. The challenges include:

- Defining abstract style concepts in ways models can understand
- Effectively disentangling stylistic elements from core action semantics and character identity
- Limited datasets with detailed style annotations

While some works aim for general motion control using interactive signals (Show Lab 2024; Kuang et al. 2024; Burgert et al. 2025), fine-grained control over expressive style remains less developed compared to action specification or identity preservation.

7.1.4 Balancing physics constraints with perceptual naturalness

The integration of explicit physical laws or biomechanical constraints into motion generation models presents a paradox. While technically correct according to physics, overly constrained motions might fall into the "uncanny valley" appearing subtly wrong or unnatural to human observers (Tevet et al. 2022). Data-driven approaches learn complex statistical regularities directly from motion capture datasets (Janson et al. 2024), capturing subtle variations and "imperfections" characteristic of real human movement. If a physics-informed model's constraints are too rigid or its integration with the learned data distribution imperfect, the resulting motion might be physically plausible but perceptually unnatural. There exists a delicate balance between enforcing explicit physical rules and capturing the nuanced patterns of biological movement learned implicitly from data. Future research must focus on achieving a harmonious blend between physics-based constraints and data-driven learning to ensure results align with human perceptual expectations of naturalness (Yan et al. 2024).

7.1.5 Identity preservation vs. motion dynamics

Achieving strong identity preservation while simultaneously generating highly dynamic and expressive motion presents an inherent tension. Techniques designed to rigidly enforce appearance consistency across frames (Xu et al. 2024; Zhang et al. 2025) can inadvertently hinder the model's ability to synthesize natural deformations, particularly in facial expressions or subtle body language. Real human motion, especially facial movement, involves complex, non-rigid changes in appearance. If a model is overly constrained to match a static reference, it may struggle to generate dynamic changes realistically, potentially resulting in a "mask-like" effect. Conversely, models allowing greater flexibility might fail to maintain identity consistency, leading to identity drift over time. Addressing this requires advancements in disentangling different aspects of human representation:

- Core, relatively static identity cues
- Underlying structure (e.g., facial landmarks, skeletal pose)
- Dynamic, time-varying expression and motion components

7.2 Future research directions

7.2.1 Technical approaches

Based on the identified research gaps, several promising research directions emerge:

1. Constraint-Aware Diffusion Models: Further development of diffusion models that intrinsically incorporate physical and biomechanical constraints. This includes refining methods using differentiable physics simulators, physics-informed neural network layers, or directly integrating physical equations into the denoising process (Yuan et al. 2023; Kang et al. 2025). Lightweight approaches capturing essential physical plausibility without full computational overhead of simulation (Tashakori et al. 2025) represent a particularly promising direction.
2. Robust Identity Preservation: Improving techniques for maintaining high-fidelity identity consistency during animation by achieving better disentanglement of static appearance features from dynamic otion and pose information (Xu et al. 2024; Zhang et al. 2025). Developing architectures robust to diverse and complex motions over long sequences, potentially leveraging training-free or few-shot adaptation methods (Show Lab 2024).
3. Nuanced Motion Style Control: Enabling explicit and fine-grained control over motion style through methods conditioning generation on textual style descriptors, learning disentangled representations separating action semantics from stylistic variations, or using reference motion clips to infer style embeddings.

4. Human-Scene Interaction Modeling: Advancing models capable of generating realistic human motions that interact plausibly with complex 3D environments. This requires models to reason about scene geometry, object properties, and physical constraints governing interactions (Yuan et al. 2023).
5. Multi-Person Interaction: Developing frameworks for modeling social dynamics and interactions between multiple human figures, accounting for interpersonal spacing, social behavior patterns, and physical contact (Tran et al. 2024).
6. Computational Efficiency: Creating optimized architectures and efficient inference methods to enable real-time or near-real-time human video generation, potentially through model compression, distillation, or specialized hardware acceleration.

7.2.2 Datasets and evaluation

Progress in human-centric motion generation depends on the availability of more comprehensive datasets and evaluation metrics:

1. Richer Datasets: Development of large-scale motion capture data annotated with detailed textual descriptions, associated physical or biomechanical measurements, information about scene context and interactions, and annotations related to motion style or affect.
2. Perceptual Evaluation Metrics: Creation of evaluation frameworks that go beyond technical accuracy to assess subjective realism and naturalness, potentially incorporating human perception studies and psychophysical measurements.
3. Benchmark Tasks: Standardization of benchmark tasks for comparing different approaches to human motion generation, with specific focus on physical plausibility, identity preservation, style control, and multi-person interaction.

7.2.3 Application domains

Human-centric video generation using diffusion models has numerous potential applications across various domains:

1. Entertainment and Creative Content: Streamlining content creation for films, games, and virtual reality experiences. Text2Performer (Jiang et al. 2023) demonstrates the potential for synthesizing video sequences from textual descriptions of appearance and motion, which could revolutionize character animation and digital human creation.
2. Fashion and Virtual Try-On: Models like DreamPose (Karras et al. 2023) designed for "generating animated fashion videos from still images" can show how clothing would look and move on different body types in various scenarios, enhancing online shopping experiences.
3. Healthcare and Rehabilitation: Generating personalized movement demonstrations for physical therapy or training, with adaptable difficulty levels and customized exercise routines.
4. Education and Training: Creating instructional videos demonstrating physical skills or procedures, potentially personalized to individual learners.
5. Social Computing: Development of more realistic and responsive avatars for virtual meetings, social platforms, and collaborative environments.

7.2.4 Ethical considerations

As human-centric video generation technology advances, important ethical considerations emerge:

1. Consent and Likeness Rights: Developing frameworks to respect individuals' rights to control the use of their likeness in generated content.
2. Misinformation Prevention: Creating built-in safeguards, watermarking, and detection mechanisms to prevent the use of this technology for creating misleading or harmful content.
3. Bias and Representation: Ensuring that generated human motions and appearances reflect the diversity of human movement patterns and do not reinforce stereotypes or biases.
4. Transparency: Establishing standards for disclosure when synthetic human videos are used in media, advertising, or other public contexts.

In the next section, we present (MIME-Vid) framework that aims at providing multi-modal human centric diffusion model framework with handling various issues were provided in this survey. Most importantly, non-realistic human motion by Kalman filter and computational complexity by quantization.

8 MIME-vid framework

8.1 Status and scope of the proposed framework

MIME-Vid is introduced as a conceptual reference framework, not as a fully-trained system with empirical benchmarks reported in this manuscript. Its role in the survey is threefold: (i) it *operationalises* the unifying principles from Sect. 4.2 (reference-flexibility, physics-perception asymmetry, and disentanglement granularity) in a single pipeline; (ii) it makes the research gaps identified in Sect. 2 concrete by specifying where Kalman filtering, quantization, and multi-modal fusion would be inserted; and (iii) it provides a testable hypothesis for future work—an empirical validation campaign, together with the evaluation protocol of Sect. 6 and the comparability caveats of Sect. 6.3, is our next step.

Accordingly, in this manuscript the novelty claim for MIME-Vid is restricted to the *integration pattern*: the combination of Kalman-filtered pose smoothing, post-training quantization, and dual-stream content guidance within a DreaMoving-family backbone has not, to our knowledge, been proposed elsewhere. Claims of quantitative superiority are deferred to a follow-up paper reporting the full empirical evaluation.

Our proposed MIME-Vid (Multi-modal Integration with Motion Enhancement for Video Generation), presents a novel enhancement to video generation systems by integrating advanced Kalman filtering techniques (Grewal 2011) with a multi-modal architecture, built upon the foundation of the DreaMoving (Feng et al. 2023) framework. MIME-Vid addresses key challenges in temporal consistency, motion realism, and computational efficiency through a carefully designed pipeline, as illustrated in the architectural diagram in Fig. 12.

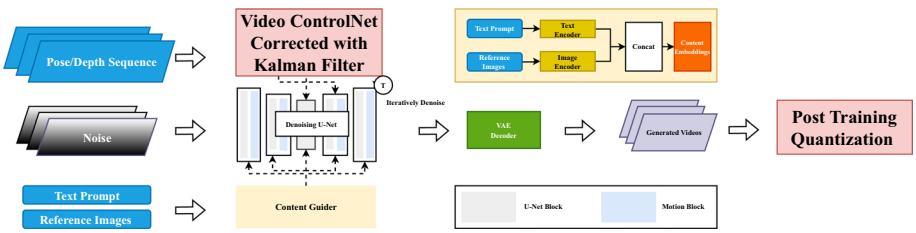


Fig. 12 MIME-Vid Framework for Video Generation with Multi-Modal Inputs: The system integrates text prompts, reference images, and pose/depth sequences through an encoder and iterative denoising process using a Denoising U-Net architecture. The framework employs a Kalman filter for temporal consistency and includes post-training quantization for efficient video generation

8.2 Multi-modal architecture analysis

We propose a novel framework architecture that will implement an advanced approach to integrating multiple input modalities while ensuring temporal coherence and motion quality. Each component has been carefully designed to address current limitations in existing video generation systems, with particular focus on human motion realism and computational efficiency. The proposed model system will processes three distinct input streams through specialized pathways. In the pose/depth sequence processing stream, raw pose and depth sequences, shown in blue in the diagram, undergo enhancement through a motion correction module incorporating Kalman filtering. This module ensures temporal consistency and natural motion transitions through integration of predictive state estimation for smooth motion. For text and image processing, MIME-Vid processes semantic descriptions and control signals alongside reference images for appearance guidance through dual-stream encoding using specialized encoders. These encoded features are then concatenated for unified representation, ultimately producing refined content embeddings for generation guidance. The noise processing pathway handles initial noise vectors for the diffusion process through structured noise handling in the denoising pipeline, providing the foundation for the diffusion-based generation process.

MIME-Vid will also implements two crucial intermediate processing modules within its pipeline. The Video ControlNet with Kalman Filter provides advanced temporal smoothing and motion prediction through sequential state estimation and correction. This implementation offers reduced motion artifacts and improved temporal coherence, with direct connection to the Denoising U-Net's motion blocks. The Content Guider implements a multi-stage processing pipeline, incorporating a text encoder for semantic feature extraction, an image encoder for visual feature processing, a concatenation layer for feature fusion, and content processing for unified representation refinement. This structure produces coherent content embeddings essential for precise generation guidance.

At the heart of our framework lies an enhanced Denoising U-Net architecture that integrates multiple sophisticated processing elements. This core component comprises a U-Net Block for primary denoising functionality, a Motion Block for specialized motion processing, and a VAE Decoder for high-fidelity video generation. The processing flow will implements iterative denoising steps, motion-aware feature processing, content-guided generation, and continuous temporal consistency maintenance throughout the generation process.

The framework generation pipeline will culminate in a sophisticated output process designed to ensure high quality video generation. This process will involve iterative refinement through denoising, VAE decoding for video synthesis, and post processing optimization. The system will maintain quality through continuous temporal consistency verification, motion quality assessment, and visual fidelity maintenance throughout the generation process.

MIME-Vid will introduce several key innovations to advance the state of video generation field. We propose integrating Kalman Filters to provide enhanced motion prediction and smoothing within the motion processing pipeline to improve temporal coherence and natural motion. The enhanced motion processing will combine U-Net and Motion Block architectures with specialized motion feature handling and temporal consistency maintenance. The system aims to achieve efficient content integration through multi-modal feature fusion, coherent content guidance, and balanced feature influence. Computational optimization will be achieved through post-training quantization, an efficient processing pipeline, and optimized resource utilization.

8.3 Integration mechanisms and module interfaces

To make the integration pattern concrete, we specify the three interfaces at which MIME-Vid's components are coupled, since it is the coupling—rather than any single module in isolation—that constitutes the proposed contribution.

(i) Kalman state: → *U-Net motion blocks.* The raw pose/depth sequence is first passed through a linear Kalman filter that maintains a per-joint state vector of position and velocity. Rather than overwriting the input pose with the filtered estimate (a hard constraint that, as in PhysDiff, tends to produce stiff motion), the filtered state

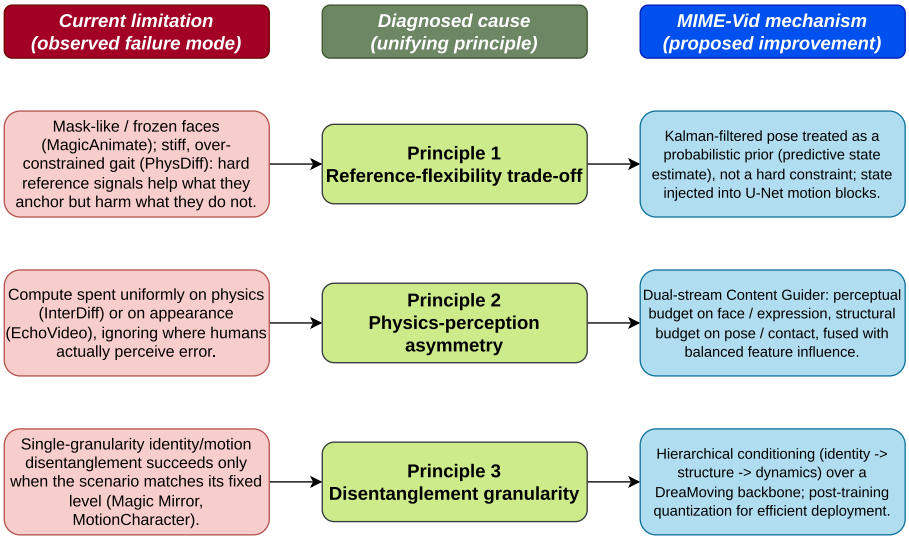


Fig. 13 From diagnosed limitations to proposed improvements. Each row links a current failure mode in human-centric video generation (left) to the unifying principle that explains it (centre, Sect. 4.2) and to the MIME-Vid integration mechanism intended to address it (right). The mapping is conceptual: it motivates the design rather than reporting trained-model results

is encoded and injected as an *additive conditioning signal* into the temporal-attention (motion) blocks of the Denoising U-Net at every denoising step. The filter's predictive estimate thus acts as a probabilistic motion prior whose influence is mediated by the network, directly operationalising Principle 1 (the reference-flexibility trade-off of Sect. 4.2).

(ii) Content Guider: → *cross-attention.* Text and reference-image features are encoded by separate encoders and fused in the Content Guider, which emits a unified content embedding consumed by the U-Net's cross-attention layers. The dual-stream design lets the appearance (image) and semantic (text) streams be weighted independently, so perceptual budget can be concentrated on identity-critical regions (face, expression) while structural budget is spent on pose and contact—an explicit realisation of Principle 2 (physics–perception asymmetry).

(iii) Post-training quantization: → *deployment.* Quantization is applied *after* training to the U-Net and decoder weights only, leaving the conditioning and fusion pathways in higher precision so that the integration behaviour above is preserved. This isolates the efficiency contribution from the quality contribution and, as committed in Sect. 6.3, allows the computational savings to be attributed to specific components in the planned ablation.

Figure 13 summarises how these mechanisms map onto the limitations they are designed to address: each row traces an observed failure mode of current methods (left), the unifying principle that explains it (centre), and the MIME-Vid mechanism that targets it (right).

9 Conclusion

Multimodal video diffusion models represent a significant step forward in the generation and control of video content. By effectively integrating various input modalities, these models enhance the ability to produce temporally coherent and semantically rich videos. Our survey has explored the foundational principles, design methodologies, and application areas of these models, highlighting how the fusion of multimodal data can improve generation quality and user control.

Despite the promising results, several challenges persist, such as improving computational efficiency, better aligning modalities, and ensuring scalability to diverse real-world scenarios. Furthermore, rigorous evaluation metrics and benchmarks are still lacking, making it difficult to compare different approaches objectively.

Future work should focus on developing more efficient and generalizable models, enhancing user interactivity, and creating standardized benchmarks. As part of fullfilling the future work, we are currently working on implementing MIME-Vid framework which is an integrated diffusion model for multi-modal human centric motion

generation with handling realistic motion by incorporating Kalman filtering and with enabling efficiency by incorporating quantization. By addressing these challenges, the field can move closer to realizing the full potential of multimodal video diffusion in practical applications.

In summary, multimodal video diffusion is a rapidly growing research area with vast potential for innovation. We hope that this survey will serve as a valuable resource for researchers and practitioners looking to advance the field further.

Acknowledgements We acknowledge the use of Claude AI (Anthropic, https://claude.ai) for assistance with writing and English proofreading. All AI-generated content was reviewed and validated by the authors.

Author contributions A.A.A. contributed to conceptualization, literature collection, methodology design, taxonomy development, data analysis, drafting of the manuscript, and preparation of figures and tables. A.R.N.-N. contributed to supervision, critical review of methodology, validation of the theoretical framework, refinement of the taxonomy, critical revisions of the manuscript, project administration, and correspondence with the journal. A.A.A. and A.R.N.-N. both contributed to the discussion of results, approved the final version of the manuscript, and agree to be accountable for all aspects of the work.

Funding The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Data availability No datasets were generated or analysed during the current study.

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